

THE FLOWERING PLANTS OF SOUTH AFRICA.

A MAGAZINE CONTAINING HAND-COLOURED FIGURES WITH DESCRIPTIONS OF THE
FLOWERING PLANTS INDIGENOUS TO SOUTH AFRICA.



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FLOWERING PLANTS OF SOUTH AFRICA; VOL. 5 ***

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THE FLOWERING PLANTS OF SOUTH AFRICA.

A MAGAZINE CONTAINING HAND-COLOURED FIGURES WITH
DESCRIPTIONS
OF THE FLOWERING PLANTS INDIGENOUS TO SOUTH AFRICA.

EDITED BY

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VOL. V.



The veld which lies so desolate and bare
Will blossom into cities white and fair,
And pinnacles will pierce the desert air,
And sparkle in the sun.

R. C. MACFIE'S "EX UNITATE VIRES."

LONDON: L. REEVE & CO., LTD., HENRIETTA STREET, COVENT
GARDEN, LONDON

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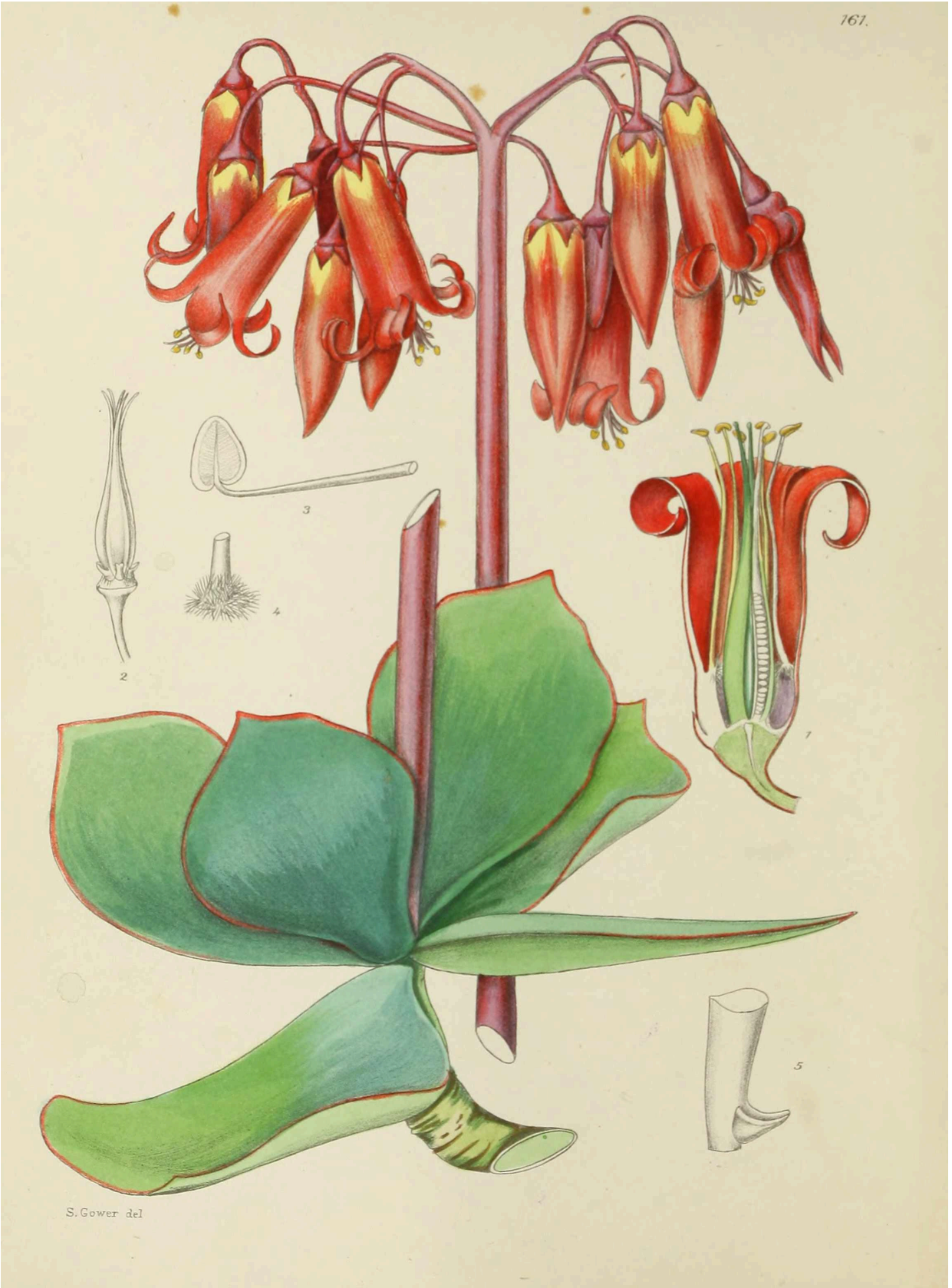
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THIS VOLUME
IS CORDIALLY DEDICATED
TO
THOMAS PEARSON STOKOE
OF CAPE TOWN

WHOSE INDEFATIGABLE EXERTIONS IN EXPLORING
THE FLORA OF THE CAPE MOUNTAINS HAVE
RENDERED VALUABLE SERVICE TO SOUTH AFRICAN
BOTANY BY THE DISCOVERY OF NEW PLANTS AND
THE RE-DISCOVERY OF MANY THAT WERE LONG
LOST, AND TO WHOSE ZEAL, ENTHUSIASM, AND
GENEROSITY AS A COLLECTOR THIS WORK IS
GREATLY INDEBTED.

DIVISION OF BOTANY, PRETORIA.

October, 1925.



S. Gower del

PLATE 161.

COTYLEDON ORBICULATA.

Cape Province, Transvaal.

CRASSULACEAE.

COTYLEDON, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. i. p. 659.

Cotyledon orbiculata, *Linn. Sp. Pl.* 614; *Fl. Cap.* vol. ii. p. 371.

Cotyledon orbiculata was first figured in the *Botanical Magazine* in 1795, and stated to have been introduced into English gardens about the year 1690. The genus *Cotyledon* differs from *Crassula* (see Plate 115) by having twice as many stamens as petals. It is a large genus in South Africa, and represented by between 30 to 40 species. Several species of the genus are of economic interest inasmuch as they produce disease in stock. The well-known "Krimptziekte" of goats is caused by *C. Wallichii*. The Division of Veterinary Education and Research carried out some feeding experiments with *C. orbiculata* in 1921 at Grahamstown, and definitely proved that feeding the leaves to fowls caused death.

Our plate was prepared from specimens grown at the Division of Botany, Pretoria. The plant goes under the common names of "hondenoor," "Konterie," "Varkens ooren," and "pig's-ear."

DESCRIPTION:—Low shrubby somewhat succulent plant. *Stem* with light-brown bark. *Leaves* opposite, 6·5 to 8 cm. long, 4 to 4·5 cm. broad, obovate, subacute, glabrous, glaucous, with red margins. *Peduncle* up to 27 cm. long, 8 mm. in diameter, terete, reddish in colour with a whitish bloom. *Inflorescence* a panicle of cymes. *Calyx-tube* almost none; lobes 5 mm. long, ovate, acute. *Corolla-tube* 2·8 cm. long, 1·5 cm. in diameter; lobes 2 cm. long, 7 mm. broad, oblong, obtuse, slightly twisted counter-clockwise. *Stamens* 10, five shorter inserted near the base of the corolla-tube, with a ring of hairs at the point of attachment and produced below the hairs into a

strong rib; filaments subterete; anthers ovate. *Carpels* as long as the shorter stamens. *Glands* forming a deep concave body at the base of each carpel.

PLATE 161.—Fig. 1, longitudinal median section of the flower; Fig. 2, gynæcium, showing glands; Fig. 3, stamen; Fig. 4, tuft of hairs on filament; Fig. 5, side view of gland at base of carpel.

F.P.S.A., 1925.



S. Cower del.

PLATE 162.

SYNNOTIA BICOLOR.

Cape Province.

IRIDACEAE. Tribe IXIEAE.

SYNNOTIA, Sweet; *Benth. et Hook. f. Gen. Plant.* vol. iii. p. 709.

Synnotia bicolor, Sweet, *Hort. Brit.* ed. 2, p. 501; *Fl. Cap.* vol. vi. p. 134.

This species was introduced into the Royal Botanic Garden at Kew by George Masson in 1786, and was described as a *Gladiolus*, and later in the *Botanical Magazine* (t. 548) as a species of *Ixia*. From the former genus it is readily distinguished by the membranous lacerated spathe-valves and from the latter by the unilateral stamens.

The plant is rather stiff, but the individual flowers are quite charming. As will be seen from the illustration, the flowers resemble in general shape those of *Gladiolus orchidi-florus* shown at Plate 165 of this work.

Our specimen was prepared from plants grown by Dr. I. B. Pole Evans, C.M.G., from corms forwarded by Mrs. E. Rood of van Rhynsdorp.

DESCRIPTION:—*Corm* ellipsoid, 3 cm. long, 1·5 cm. in diameter, covered with fine reticulated sheaths. *Leaves* 7 in a basal distichous rosette, 8 to 11 cm. long, 0·7 to 1·4 cm. broad, linear-oblong, very acute, with 1 more or less evident mid-rib, but many-veined when viewed in transmitted light. *Peduncle* about 12 cm. long, flexuose, bearing about 6 distant flowers. *Outer spathe-valves* 1 cm. long, deeply 3-partite, inner 2-partite. *Perianth* distinctly 2-lipped; tube 1·5 cm. long, widening upwards; posterior segment 2·5 cm. long, erect, clawed, with an ovate obtuse limb; side-segments 1·7 cm. long, 6 mm. broad, more or less oblong, obtuse, spreading-reflexed; 3 anterior segments more or less horizontal and forming a distinct lip. *Style* as long as the stamens, divided into 3 lobes dilated at the apex. (National Herb. Pretoria, No. 2860.)

PLATE 162.—Fig. 1, median longitudinal section of flower; Fig. 2, spathe-valve; Fig. 3, stamen; Fig. 4, style showing stigmas; Fig. 5, fruit.

F.P.S.A., 1925.



S. Gower del.

PLATE 163.

GLADIOLUS TRICHOSTACHYS.

Transvaal.

IRIDACEAE. Tribe IXIEAE.

GLADIOLUS, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 709.

Gladiolus trichostachys, *Baker in Bull. Herb. Boiss.* ser. II. vol. iv.
p. 1006.

This charming little *Gladiolus* was first found by Conrath at Irene, near Pretoria, and was again collected by Dr. I. B. Pole Evans, C.M.G., in the same locality. It was described by Baker, with several other Transvaal species, in 1904, but is here figured for the first time.

The plant produces a single erect flowering stem with clasping leaves scarcely produced above. The species belongs to the Section *Hebea*, which we have illustrated on Plates 63 and 165 (*G. alatus*, *G. orchidiflorus*), and is closely related to *G. permeabilis*, a species common in the Cape Province, but which also extends into Bechuanaland. It is, however, easily distinguished from this species by the non-produced leaves and hairy stems.

Our illustration was made from the specimens collected by Dr. I. B. Pole Evans, C.M.G., at Irene.

DESCRIPTION:—*Bulb* 2 cm. in diameter, globose, covered with brown fibres. *Stem* pilose on the lower half. *Leaves* clasping the stem, sheath pilose, hardly produced. *Inflorescence* 13 cm. long, 7-to 10-flowered. *Outer spathe-valves* 1·5 cm. long, somewhat membranous above, entire, bifid or trifid; inner spathe-valves similar to the outer, bifid. *Stamens* shorter than the style; anthers somewhat sagittate at the base. *Style-branches* cuneate, fimbriate on the edges.

PLATE 163.—Fig. 1, plant much reduced;
Fig. 2, median longitudinal section of flower;
Fig. 3, outer spathe-valve; Fig. 4, inner

spathe-valve; Fig. 5, anther; Fig. 6, portion of style with style branches.

F.P.S.A., 1925.

Sect above base
showing indentations



PLATE 164.

ALOE CHABAUDII.

Rhodesia.

LILIACEAE. Tribe ALOINEAE.

ALOE, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 776.

Aloe Chabaudii, *Schonl. in Gard. Chron.* 1905, p. 162.

Dr. Schonland in his description of this *Aloe* in the *Gardeners' Chronicle* states that it is allied to *A. striata*, and a reference to our Plate 55 will show the similarity of the flowers in the two species. The specimens from which the original description was prepared were collected by Mr. J. M. Brown in Rhodesia, but the exact locality is not known. They were grown and flowered by Mr. J. A. Chabaud of Port Elizabeth. The plant is erect with a short stem, and has not the reclining habit of *A. striata*. It is very suitable for large rockeries, and makes an effective display when in flower.

Our plate was prepared from specimens grown at the Division of Botany, Pretoria.

DESCRIPTION:—Acaulescent or almost so. *Leaves* 18 to 24, forming an irregular rosette, up to 45 cm. long and 15 cm. broad near the base, about 2 cm. thick, ovate-lanceolate, unspotted, somewhat glaucous; upper surface indistinctly striate, nearly flat, except near the apex, where it is channelled; lower surface slightly convex; margin with a narrow horny border; prickles 1·5 mm. long or even smaller, at first flesh-coloured, brown in older leaves, straight or (especially in the upper portion of the leaf) curved forward, about 1·5 cm. apart, interspaces straight. *Inflorescence* a loose panicle, with squarrose ascending branches 60 to 80 cm. in height, about 45 cm. in diameter; racemes lax, floriferous portion 15 to 20 cm. long; bracts deltoid, acuminate, membranous, lowest about 6 mm. long, upper gradually smaller; pedicels spreading, lowest 2 cm. long, upper only slightly smaller; perianth 3·5 cm. long, slightly curved, distinctly obconical at the base, swollen round the ovary, with three decided oblong indentations above it (in a line with the

inner segments); outer segments pale brick-red with nearly white wings near the apex, inner with red median line and pale wings, which are yellowish at the apex; tube of corolla nearly two-thirds its length; filaments yellow, slightly exceeding the perianth in length, anthers pale terra-cotta; ovary broadly oblong, green; style yellow, not exserted; stigma very small, capitate (S. Schonland).

PLATE 164.—Fig. 1, median longitudinal section of flower; Fig. 2, bract; Fig. 3, anther; Fig. 4, apex of style.

F.P.S.A., 1925.



S. Gower del.

PLATE 165.

GLADIOLUS ORCHIDIFLORUS.

Cape Province.

IRIDACEAE. Tribe IXIEAE.

GLADIOLUS, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 709.

Gladiolus orchidiflorus *Andr. Bot. Rep.* t. 241; *Fl. Cap.* vol. vi. p. 160.

In the *Botanical Magazine* for the year 1803 (Plate 688) appeared an excellent figure of this species under the name *Gladiolus viperatus*, which was given owing to a fanciful resemblance to “the head and appearance of the jaws of the snake when raising itself for defence against its enemy and hissing.” Prior to the figure above quoted Jacquin illustrated the same species (1781-1786).

Like most species of *Gladiolus*, it lends itself to cultivation and is easily grown in pots.

A comparison of this plate of *G. orchidiflorus* and Plate 63 (*G. alatus* var. *namaquensis*) should be made with other plates of *Gladiolus* figured. The two species belong to the section *Hebea*, all the members of which may be easily recognised by the long claws to the perianth-segments.

The plants from which our illustration was made were grown by Dr. I. B. Pole Evans, C.M.G., at Irene, near Pretoria, from corms sent by Mrs. E. Rood of van Rhynsdorp.

DESCRIPTION:—*Corm* 2·5 cm. diameter, depressed-globose, covered with fibrous tunics. *Produced leaves* 4; lowest leaf with a lanceolate-linear blade, 6·5 cm. long, 8 mm. broad, prominently 2-ribbed; upper leaves up to 30 cm. long, 5 to 8 mm. broad, linear, acuminate, acute, with 2 of the ribs more prominent than the others, glabrous. *Inflorescence* about half as long as the leaves, laxly few-flowered. *Perianth* very unequal; the uppermost segment

long-clawed and with an oblong obtuse blade membranous on the margins, arched over the flower and the style and stamens; side segments broadly ovate, clawed, and produced into an acuminate point; lower segments more or less forming a lip, spathulate, long-clawed, obtuse. *Stamens* arched under the uppermost segment and completely hidden by it. *Style* arched under the uppermost segment, projecting beyond its apex; lobes spathulate, papillose round the edges. (National Herb. Pretoria, No. 2858.)

PLATE 165.—Fig. 1, plant much reduced; Fig. 2, median longitudinal section of flower; Fig. 3, corm; Fig. 4, spathe-valve; Fig. 5, stamen; Fig. 6, top of style showing the 3 stigmas; Fig. 7, fruit.

F.P.S.A., 1925.



S. Gower del.

PLATE 166.

LACHENALIA TRICOLOR.
var. LUTEOLA.

Cape Province.

LILIACEAE. Tribe SCILLEAE.

LACHENALIA, *Jacq.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 807.

Lachenalia tricolor, *Thunb.* var. **luteola**, *Baker.; Jacq. Collect.* vol. iv.
p. 148; *Ic.* vol. ii. p. 16, t. 395; *Fl. Cap.* vol. vi. p. 424.

The *Lachenalia* figured on the accompanying plate is among the most graceful species in the genus. It is closely allied to *L. pendula*, illustrated on Plate 158, but is distinguished by having the inner perianth-segments much longer than the outer. The plant was known to horticulturists in Europe almost 150 years ago, and was figured in colour between 1786 and 1793 by Jacquin, and again in the *Botanical Magazine* in 1807.

L. tricolor grows in the sandy parts of the Cape Province, but lends itself to cultivation in pots. It has been successfully grown at Irene, near Pretoria, by Dr. I. B. Pole Evans, C.M.G., from bulbs supplied by Lady Smartt of "Glen Ban," Stellenbosch, C.P., and from these specimens our illustration was made.

In the young flowering stage the buds are quite green, but become yellow in the lower half as they grow older, while in the adult flower the colour is a deep chrome (R. C. S.).

DESCRIPTION:—*Bulb* globose, 1·3 cm. in diameter, with long white roots from the base. *Leaves* 2, up to 17 cm. long, 2·5 cm. broad at the base, 2 cm. broad above, strap-shaped, obtuse, glabrous. *Peduncle* up to 16 cm. long, terete, green, spotted with brown. *Flowers* racemose, pendulous, each flower arising from a small pocket formed by the bract. *Bract* 3 mm. long, ovate-

lanceolate, obtuse, with a conical blunt spur. *Outer perianth-segments* 1·6 cm. long, oblong, obtuse, one slightly beaked below the apex; inner segments 2·6 cm. long, widened upwards, obtuse. Stamens hardly exerted. *Style* as long as the stamens, penicillate at the apex. (National Herb. Pretoria, No. 2857.)

PLATE 166.—Fig. 1, whole plant much reduced; Fig. 2, median longitudinal section of the flower; Fig. 3, bulb; Fig. 4, part of peduncle showing pocket-like bracts; Fig. 5, stamen; Fig. 6, upper portion of style showing stigma.

F.P.S.A., 1925.



S. Gower del.

PLATE 167.

CRASSULA ROSULARIS.

Cape Province, Natal.

CRASSULACEAE.

CRASSULA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. i. p. 657.

Crassula rosularis, *Harv.; Fl. Cap.* vol. ii. p. 350.

Crassula rosularis belongs to the section *Rosulares*, which is characterised by having rosulate flat radical leaves and a scape-like flowering stem. Unlike so many species of *Crassula*, it is a shade-loving plant.

The plant from which our illustration was made was found growing under Aloes near Greytown in Natal, whence it extends southwards to Uitenhage. It is a dainty little plant, and would thrive on a shaded rockery if supplied with humus and a fair amount of moisture. In its choice of habitat it resembles very much *C. flabellifolia*, *C. Saxifraga* and *C. Septas*.

We are indebted to Lady Leuchars for the specimens.

DESCRIPTION:—An acaulescent plant. *Leaves* rosulate, radical; lower leaves spatulate-oblong, about 5 cm. long; the upper leaves becoming gradually smaller and broadly ovate; all with cartilaginous-ciliated margins. *Peduncle* scape-like, glabrous. *Flowers* in peduncled cymes, arranged in a panicle on a common peduncle. *Pedicels* about 2 mm. long. *Calyx* half as long as the corolla; lobes lanceolate-oblong, ciliate. *Petals* obovate-oblong, with a dorsal apiculus just below the apex. *Stamens* 5, almost as long as the petals and alternating with them. *Glands* of 5 scales opposite the carpels. *Carpels* 5; styles short; stigmas capitate. (National Herb. Pretoria, No. 2859.)

PLATE 167.—Fig. 1, median longitudinal section of flower; Fig. 2, carpels; Fig. 3, stamens.

F.P.S.A., 1925.



S. Gower del.

See leaf on original, this gives
wrong impression.

PLATE 168.

LISSOCHILUS SPECIOSUS.

Cape Province, Natal, Transvaal.

ORCHIDACEAE. Tribe Vandeeae.

LISSOCHILUS, *R. Br.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 536.

Lissochilus speciosus, *R. Br. ex Lindl. Coll. Bot.* t. 31; *Fl. Cap.* vol. v. sect. iii. p. 59.

Lissochilus speciosus is one of the more common species of the genus and has a wide range of distribution. It is found in the Uitenhage Division, and then follows the coastal belt northwards, through the Transkei into Natal, and up to the northern spurs of the Drakensbergen in the Transvaal. The species has also been recorded from Mazoe in Rhodesia. Robert Brown's genus *Lissochilus*, which he founded in 1821, was based on this species. But the late Dr. H. Bolus placed it in the genus *Eulophia*, and redescribed the plant as *Eulophia speciosa* in 1890, and figured it under the same name in 1911 (*Orchids of South Africa*, vol. ii. t. 13).

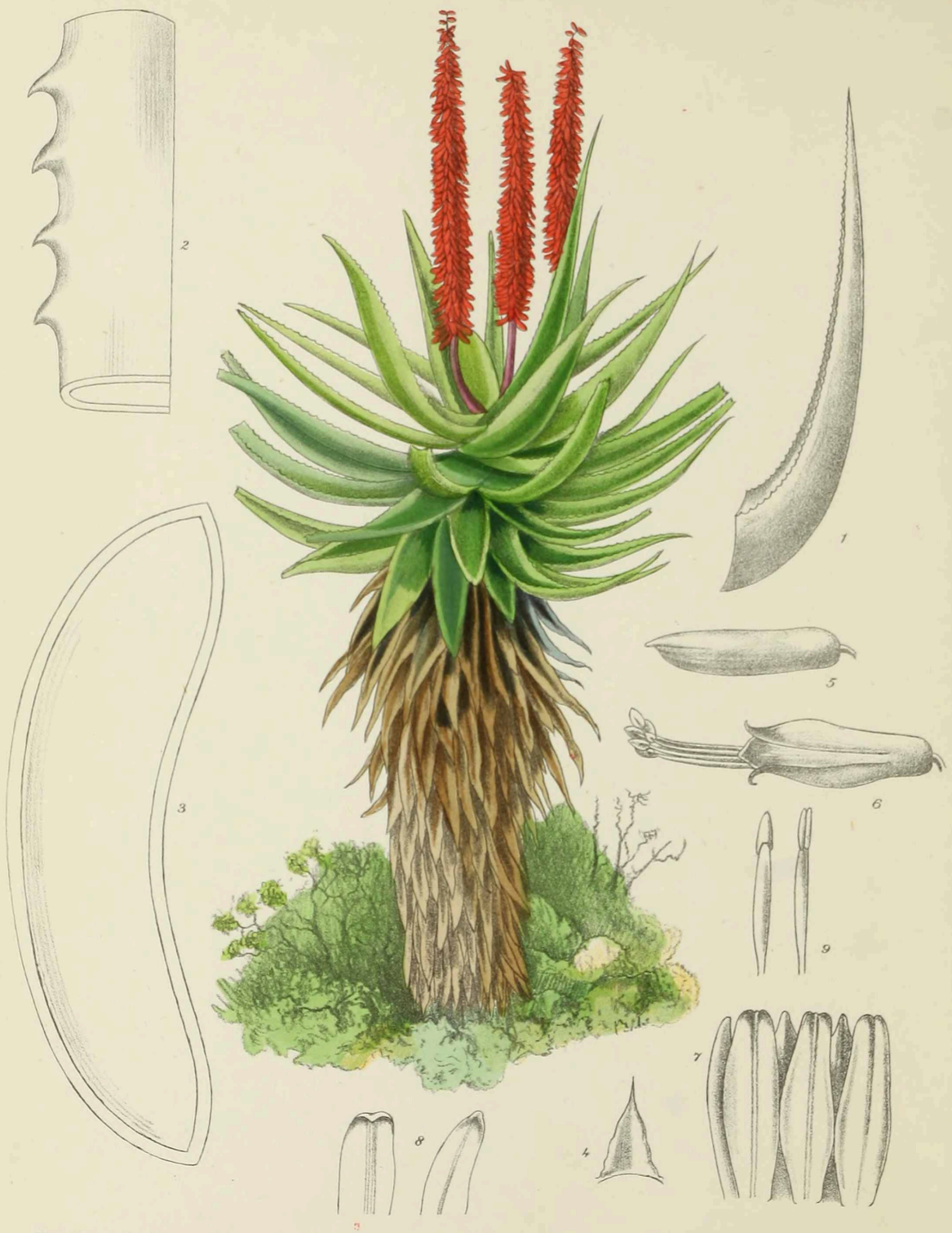
The specimen from which our plate was made was found by Misses H. Forbes and S. Gower at Isipingo, Natal, and grown at the Division of Botany, Pretoria. It is common all along the slopes of the sandhills of the Southern Natal coast.

DESCRIPTION:—Pseudobulbs ovoid, 5 cm. or more long, with a few ovate sheaths, 3-to 5-leaved; leaves elongate, linear, acute, somewhat fleshy, without prominent veins, conduplicate below, not articulated above the base, 15 to 30 cm. or more long, 2 to 2·5 cm. or more broad; scapes erect, stout, up to nearly 1 m. long, with several spathaceous sheaths; racemes long, somewhat lax, many flowered, flowers medium-sized, bracts ovate-oblong to ovate-lanceolate, acuminate, 2 to 2·5 cm. long; pedicels 2 to 2·5 cm. long; sepals ovate to ovate-oblong, subacute or acute, reflexed, green, about 1 cm. long; petals spreading, broadly ovate or ovate-suborbicular, subobtusely, about 2 cm. long, bright yellow; lip 3-lobed, nearly as long as the petals; side-lobes

suberect, short and transversely oblong, white with a few reddish lines; front lobe broadly elliptic, obtuse, reflexed at the sides, yellow with a few reddish lines at the base; disc convex, with 3 obtuse keels; spur very short, broadly conical, obtuse, column oblong, 6 mm. long (*Flora Capensis*).

PLATE 168.—Fig. 1, plant much reduced;
Fig. 2, median longitudinal section of flower;
Fig. 3, bract; Fig. 4, anthers; Fig. 5, back view
of anthers.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 169.

ALOE FEROX.

Cape Province, Natal, Transvaal.

LILIACEAE. Tribe ALOINEAE.

ALOE, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 776.

Aloe ferox, *Miller, Gard. Dict.* ed. viii. No. 22; *Fl. Cap.* vol. vi. p. 326.

This species of *Aloe* is unique among the South African representatives of the genus as being of some economic importance. The thick juice of the leaves yields the commercial product “aloes.” The method of preparing “aloes” is as follows: A hole is scooped in the ground and lined with a skin, and the cut ends of the leaves are placed on the skin so that the juice exudes and collects. The thick juice is then heated, and on cooling the “aloes” crystallise out.

Aloe ferox is very common in parts of the south-eastern Cape Province and in the midlands of Natal, and the plants form a very characteristic feature in the landscape. Plants may reach a height of 8 to 12 feet, and the simple stem is crowned with a dense rosette of leaves, while the lower portion of the stem is covered with the remains of the leaves.

Our plate was prepared from a specimen flowering at the Division of Botany, Pretoria.

DESCRIPTION:—*Stem* simple. *Leaves* many in a dense terminal rosette, varying from 0·5 to 1 m. long, 10 to 15 cm. broad below and gradually narrowing above, convex on the lower surface, concave on the upper surface, prickly on the edges; prickles stout, slightly recurved. *Inflorescence* a terminal branched raceme. *Racemes* up to 0·6 m. long, very dense. *Bracts* ovate, cuspidate. *Perianth-segments* 1·3 cm. long, 8 mm. broad, oblong,

cucullate at the apex. *Filaments* linear; anthers not much broader than the filaments. *Style* cylindric, stigma simple.

PLATE 169.—Fig. 1, leaf ($\times 1/8$); Fig. 2, margin of leaf showing prickles; Fig. 3, cross-section of leaf; Fig. 4, bract; Fig. 5, flower bud; Fig. 6, mature flower; Fig. 7, perianth-segments; Fig. 8, portion of inner and outer perianth-segments; Fig. 9, stamens, front and back view.

F.P.S.A., 1925.

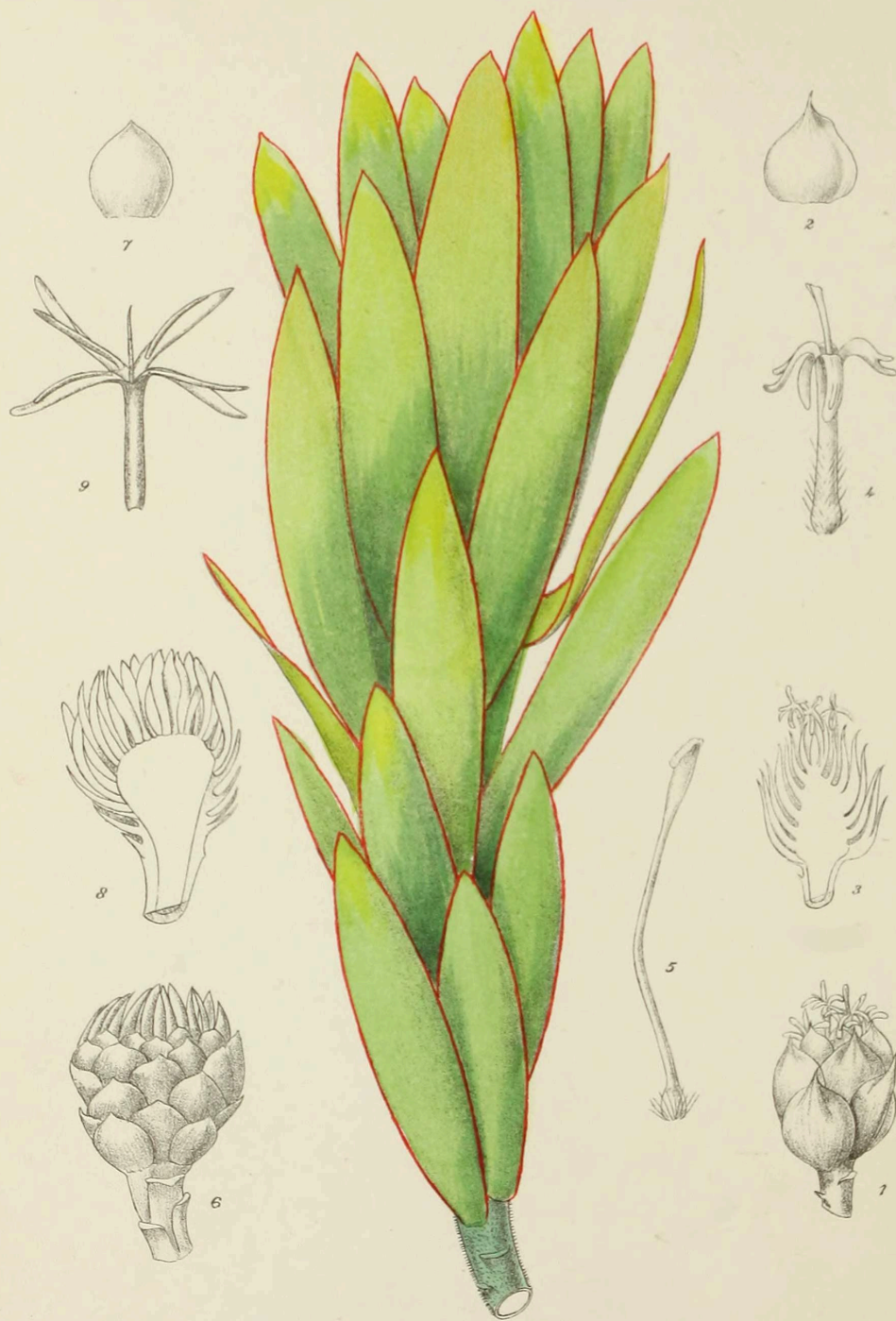


PLATE 170.

LEUCADENDRON HUMIFUSUM.

Cape Province.

PROTEACEAE. Tribe PROTEAE.

LEUCADENDRON, *R. Br.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 169.

Leucadendron humifusum, *E. Mey. in Drege, Zwei. Pfl. Documente*,
pp. 64, 118, 198; *Fl. Cap.* vol. v. sect. i. p. 549.

This interesting species of *Leucadendron* was first collected by Drege between the years 1826 and 1829, and then completely lost sight of, as none of the later collectors are credited with finding it again. In August, 1922, Mr. T. P. Stokoe came across the plant on the Hottentot Holland Mountains, and thus had the honour of bringing to the notice of South African botanists a species which had not been collected for almost one hundred years. We take this opportunity of giving a fuller description of the plant than that appearing in the *Flora Capensis*, and describe the female plant for the first time.

DESCRIPTION:—A shrub. *Branches* tomentose, at length becoming glabrous. *Leaves* 3·5 to 7 cm. long, 0·7 to 1·6 cm. broad (those surrounding the heads a little larger), oblong or oblong-lanceolate, with a blunt callous apex, slightly narrowed to a rather broad base, with 3 distinct veins from above the base, glabrous. *Male head* sessile, 2·5 to 3 cm. long (including the flowers), 2·5 cm. in diameter, surrounded by about 8 series of involucral bracts. *Involucral bracts* 1·15 to 1·3 cm. long, 8 to 8·5 mm. broad, oblong, the outer shortly cuspidate and with ciliated margins, inner rounded at the apex and without cilia, all glabrous. *Receptacle* 1 cm. high, 1 cm. in diameter, club-shaped. *Perianth-tube* 7 mm. long, somewhat compressed, glabrous; lobes 8 mm. long, linear, obtuse at the apex, glabrous. *Anthers* 5 mm. long, linear. *Style* 1·1 cm. long, terete, pilose below; stigma faintly two-lobed. *Female-head* 1·6 cm. long, 2 cm. in diameter, surrounded by about 4 series of involucral bracts. *Involucral bracts* 1·2 to 1·5 cm. long, up to 1·4 cm. broad, ovate, shortly cuspidate, obtuse, the outer ciliated, the inner

without cilia, all glabrous. *Receptacle* 1 cm. high, 6 mm. broad, conical. *Perianth-tube* 8 mm. long, compressed, long-pilose; limb 2 mm. long, linear, obtuse, glabrous. *Staminodes* 0·75 mm. long. *Ovary* 1 mm. long, 0·75 mm. in diameter, ellipsoid, pilose; style 1 cm. long, linear, gradually narrowing to the base; stigma flat and oblique.

PLATE 170.—Fig. 1, female head; Fig. 2, bract; Fig. 3, longitudinal section of female head; Fig. 4, female flower; Fig. 5, pistil; Fig. 6, male head; Fig. 7, bract; Fig. 8, longitudinal section of male head; Fig. 9, male flower.

F.P.S.A., 1925.



S. Gower del.

PLATE 171.

ALOE MARLOTHII.

Transvaal.

LILACEAE. Tribe ALOINEAE.

ALOE, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 776.

Aloe Marlothii, *Berger in Engl. Bot. Jahrb.* vol. xxxviii. p. 87.

Aloe Marlothii is a very close ally of *A. ferox* figured on Plate 169, but a careful comparison of the two will show points of difference which enables one to distinguish the two species. The inflorescence in *A. Marlothii* has the primary branches horizontal and the flowers are more or less on one side of the axis, and not arranged so as to form a cylindric raceme, as in *A. ferox*. The leaves of the species also differ, those of *A. Marlothii* being concave-convex in cross-section and with both the upper and lower surfaces bearing prickles, while in *A. ferox* the leaf is biconvex in cross-section.

Plants of *A. Marlothii* up to 15 ft. high are often found, and they make a very ornamental show in the rockery.

Our plate was prepared from specimens grown at the Division of Botany, Pretoria.

DESCRIPTION:—*Leaves* up to 1 m. long, ovate-lanceolate, acuminate, acute, concave-convex in cross-section, prickly on the face and back, with the margins armed with prickles. *Inflorescence* a branched raceme, with the primary branches horizontal. *Flowers* more or less secund. *Bracts* broadly ovate, shortly acuminate. *Perianth* tubular. *Stamens* at length exserted. *Ovary* ellipsoid, style cylindric, at length exserted; stigma small.

PLATE 171.—Fig. 1, plant, much reduced;
Fig. 2, median longitudinal section of flower;
Fig. 3, bract; Fig. 4, part of stamen; Fig. 5,
pistil.

F.P.S.A., 1925.



S. Gower del.

PLATE 172.

HYPOXIS ROOPERI.

Basutoland, Cape Province, Natal, Transvaal.

AMARYLLIDACEAE. Tribe HYPOXIDEAE.

HYPOXIS, Linn.; *Benth. et Hook. f. Gen. Plant.* vol. iii. p. 717.

Hypoxis Rooperi, *Moore in Gard. Comp.* 1, 65, cum icone; *Fl. Cap.*
vol. vi. p. 188.

The accompanying plate is our first illustration of a characteristic South African genus, namely, *Hypoxis*. The genus contains over sixty species, of which number more than forty are found in the Union, the remainder being natives of tropical Africa, tropical Asia, Australia and America.

The species is acaulescent, with a large underground corm crowned with a ring of bristles, and bears a number of distichous leaves. The flower-stalks arise from the axils of the leaves.

It is quite a common plant in the south-eastern portion of the Cape Province, and extends through East Griqualand into Natal and northwards into the Drakensbergen round Barberton. It has, however, also been recorded from the Potchefstroom District in the Transvaal.

In winter the leaves die down, but appear again as soon as the frosts cease, and sometimes long before the summer rains commence the plants are in full bloom in the veld.

Our plate was prepared from specimens flowering at the Division of Botany, Pretoria.

DESCRIPTION:—*Leaves* up to 30 cm. long, 3·2 cm. broad, strap-shaped, narrowing upwards, falcate, folded from the midrib, closely and distinctly ribbed, almost glabrous on the upper surface, softly pilose on the back and margins. *Peduncles* much shorter than the leaves, villous. *Bracts* 1·3 cm.

long, linear, acute, villous on the back. *Pedicels* up to 1·6 cm. long, villous. *Outer perianth-segments* 2·2 cm. long, 8 mm. broad, oblong, obtuse; inner segments 2·1 cm. long, 1·5 cm. broad, elliptic, obtuse. *Filaments* shorter than the anthers. *Ovary* subglobose, villous, stigmas 3-lobed, papillose on the margins.

PLATE 172.—Fig. 1, plant, much reduced; Fig. 2, portion of peduncle showing bracts and part of pedicels; Fig. 3, flower, surface view; Fig. 4, flower, back view; Fig. 5, anthers; Fig. 6, stigma, side and top views; Fig. 7, transverse section of ovary.

F.P.S.A., 1925.

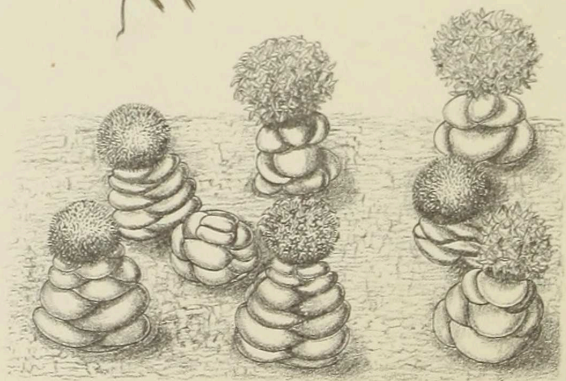
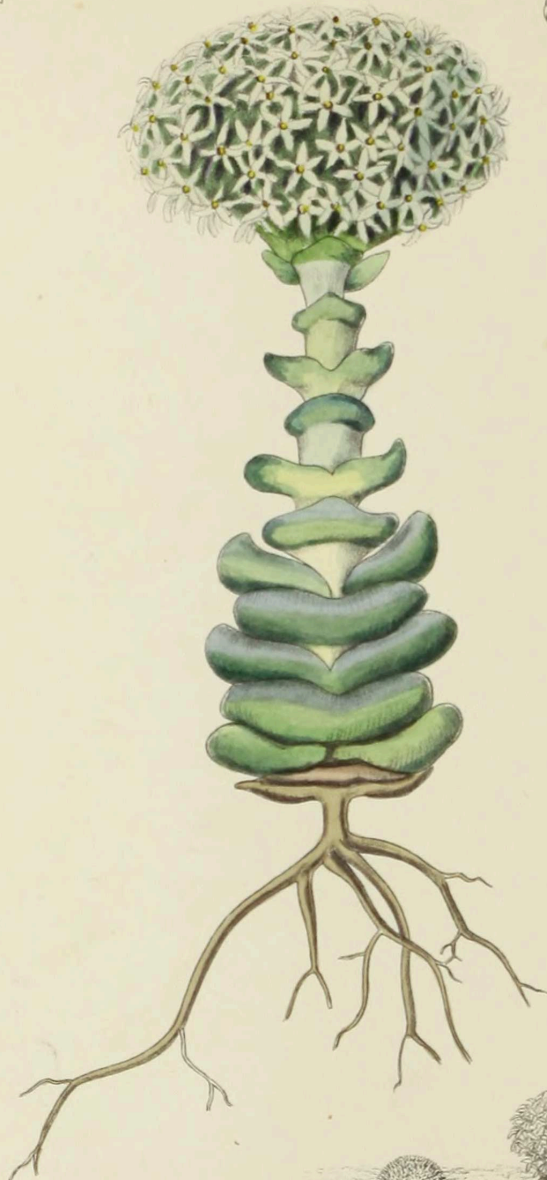
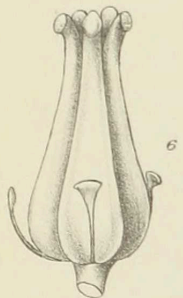
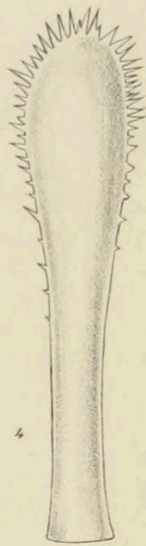
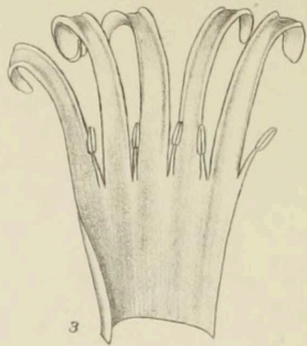
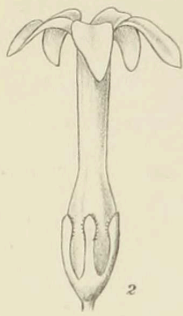


PLATE 173.

CRASSULA COLUMNARIS.

Cape Province.

CRASSULACEAE.

CRASSULA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. i. p. 657.

Crassula columnaris, *Linn. f. Suppl.* 191; *Fl. Cap.* vol. ii. p. 358.

This quaint little *Crassula* may be found in parts of the Karroo and also in Namaqualand. It has on several occasions been collected in the neighbourhood of Matjesfontein and Ceres. In its native home, where it is usually exposed to the full force of the sun's rays, it is far more compact than is shown in our illustration, which was made from a plant grown partly in the shelter of a tree. It is easily cultivated on the rockery if not kept too moist, as an abundance of water causes the plant to rot. It is a charming object when in flower, and growers of South African succulents should certainly try to secure specimens of this species, as in shape it is more or less unique in the genus *Crassula*.

Our plate was prepared from specimens presented by Mr. A. J. Austin of Matjesfontein, and grown at the Division of Botany, Pretoria. We are indebted to Dr. R. Marloth for a photograph of the plant as it grows, and part of the plate has been prepared from this.

DESCRIPTION:—*Leaves* 2·8 cm. broad, usually less than 1 cm. long, deeply concave on the inner face. *Inflorescence* a compact globose head. *Calyx* shortly campanulate at the base; lobes linear-spathulate, fringed above with papillose hairs. *Corolla* somewhat ventricose below, tubular above; lobes linear-spathulate. *Stamens* much shorter than the corolla-lobes. *Hypogynous glands* spathulate above, narrowed into a long claw. *Carpels* ventricose below, narrowed upwards, somewhat recurved above.

PLATE 173.—Fig. 1, a single leaf and cross-section of leaf; Fig. 2, single flower; Fig. 3, corolla laid open showing the stamens; Fig. 4, calyx-lobe, much enlarged; Fig. 5, corolla-lobe, enlarged; Fig. 6, gynaecium, showing hypogynous glands.

F.P.S.A., 1925.



K. A. Lansdell del.

PLATE 174.

SENECIO TAMOIDES.

Cape Province, Natal, Transvaal.

COMPOSITAE. Tribe SENECONIDEAE.

SENECIO, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. ii. p. 446.

Senecis tamoides, *DC. Prodr.* vol. vi. p. 403; *Fl. Cap.* vol. iii. p. 404.

The species here figured belongs to a small group of species in the genus (Section *Scandentes*), which is characterised by its members being climbing, half-climbing or trailing plants. *Senecio tamoides* is a very common plant in the bush and the coastal belt, and its large trusses of bright yellow, honey-scented flowers show up in strong contrast against the background of dark green foliage. At Durban, Natal, the flowers are frequently attacked by an insect, and as a result a gall is formed inside the flower head. The gall enlarges into a cylindric green body much longer than the flower-head, and each contains a single larva.

Senecio tamoides has been recorded from the Chipete Forest in Rhodesia, and was also collected by Mr. E. E. Galpin, F.L.S., at Barberton in the Transvaal.

Our illustration was made from specimens collected by Miss K. A. Lansdell on the Berea, Durban.

DESCRIPTION:—A climbing herb. *Branches* glabrous. *Leaves* 2·2 to 5·5 cm. long, 2·7 to 6 cm. broad, ovate, acuminate, subobtusate, somewhat hastate, with the margins acutely lobulate, glabrous. *Inflorescence* a many-headed corymb. *Involucral-bracts* about 6, 8 mm. long, 2 mm. broad, oblong-linear, obtuse, with membranous margins. *Ray-florets*: *Tube* 5 mm. long, cylindric; *limb* 6·5 mm. long, 2 mm. broad, oblong. *Style* exserted; style branches filiform. *Disc-florets* male, cylindric, 6·5 mm. long, lobes ·75 mm. long, linear, obtuse. *Anthers* blunt. *Style-branches* linear, truncate. *Ovary* infertile, 3 mm. long, cylindric, ribbed, glabrous. *Pappus* copious.

PLATE 174.—Fig. 1, ray-floret; Fig. 2, disc-floret.

F.P.S.A., 1925.



PLATE 175.

GLADIOLUS TRISTIS.

Cape Province.

LILIACEAE. Tribe IXIEAE.

GLADIOLUS, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 709.

Gladiolus tristis, *Linn. Sp. Plant.* ed. 2, i. 53, ex parte; *Fl. Cap.*
vol. vi. p. 139.

There appears to be a great deal of confusion between this species of *Gladiolus* and *G. grandis*, even among the specimens quoted under these species in the "Flora Capensis"; and possibly when a large range of living specimens is examined the two species now upheld will be referred to a single species. *Gladiolus tristis* is quite a handsome member of the genus, and while not so highly coloured as some, the large semi-translucent flowers with such delicate markings have a charm of their own; besides which, the flowers are very sweetly scented. The species is readily raised from seed, and under proper cultivation it flowers about fifteen months after sowing.

Our plate was made from plants grown by Dr. I. B. Pole Evans, C.M.G., at Irene near Pretoria.

DESCRIPTION:—*Corm* 1·5 cm. in diameter, globose. *Produced leaves* 2 or 3; the lowest up to 36 cm. long, 4-angled, appearing as a cross in transverse section, glabrous; upper leaves similar but shorter. *Inflorescence* usually 3-flowered. *Outer spathe-valves* 3·5 cm. long, lanceolate-oblong, acute; inner spathe-valve similar, but slightly curved and gradually narrowing upwards; lobes 3·5 cm. long, 2·8 cm. broad, ovate, bluntly acuminate, obtuse minutely apiculate. *Stamens* shorter than the style. *Style-branches* cuneate, papillose on the margins.

PLATE 175.—Fig. 1, plant, much reduced;
Fig. 2, corm, showing bulbil; Fig. 3, median
longitudinal section of flower; Fig. 4, portion

of leaf; Fig. 5, cross-section of leaf; Fig. 6, anther with part of filament; Fig. 7, stigmae with part of style; Fig. 8, ovary.

F.P.S.A., 1925.

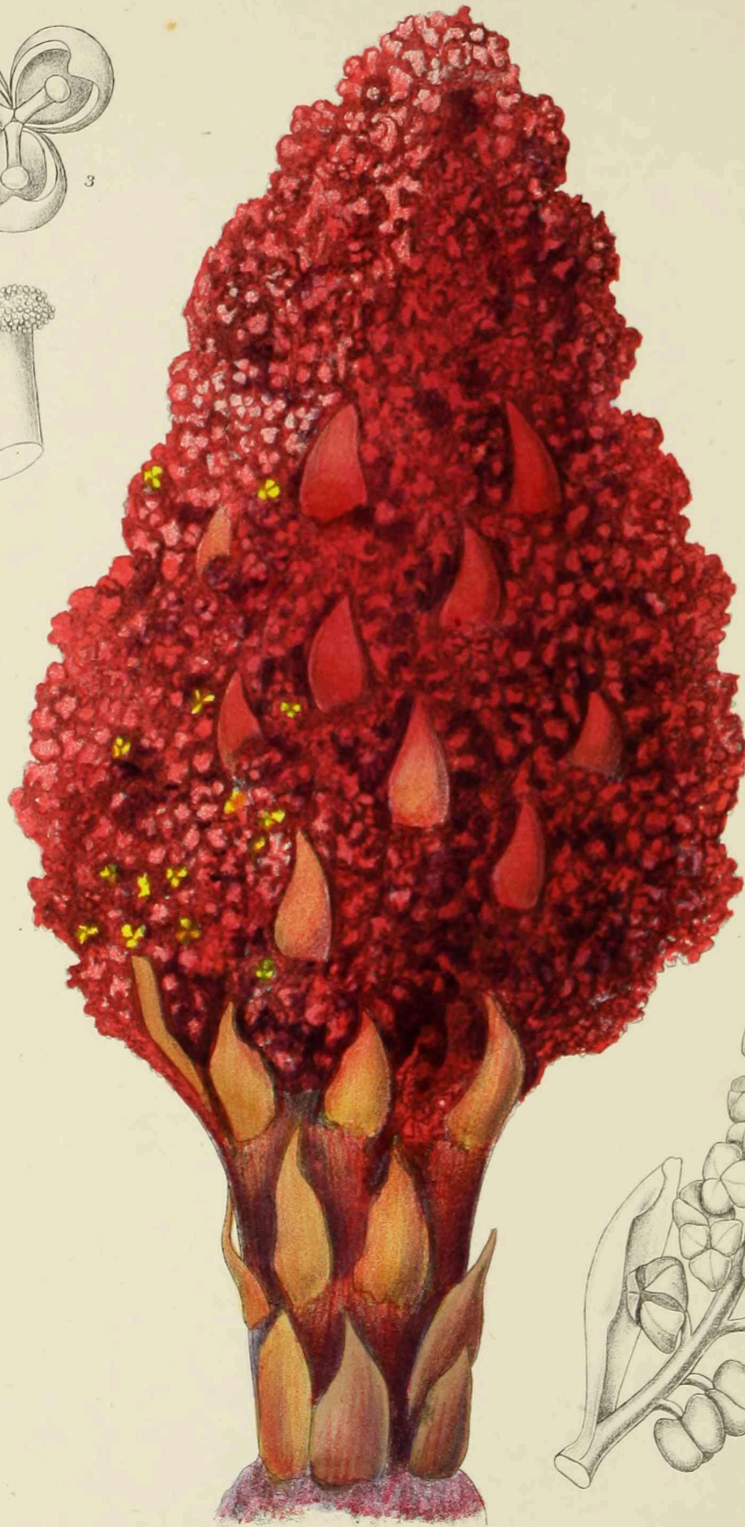
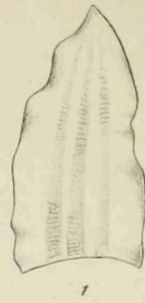
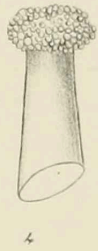
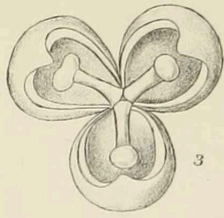


PLATE 176.

SARCOPHYTE SANGUINEA.

Cape Province.

BALANOPHORACEAE. Tribe SARCOPHYTEAE.

SARCOPHYTE, *Sparrm.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 234.

Sarcophyte sanguinea, *Sparrm. in Vet. Acad. Handl. Stockh.* 1776, 300, t. 7; *Fl. Cap.* vol. v. sect. ii. p. 213.

The family *Balanophoraceae*, to which our plant belongs, comprises plants all of which are parasitic on the roots of trees and shrubs. There are about fifty species, spread over about fifteen genera, widely distributed in tropical and sub-tropical regions. In South Africa the family is represented by two genera and four species.

Sarcophyte sanguinea is usually found as a parasite on the roots of *Acacia Karroo*, but so far as we know does not follow the same distribution as its host, but is confined to the eastern portion of the Cape Province. The plant really consists of a large inflorescence arising from the ground, and the two sexes are distinct. When in flower it has a most objectionable stench, so much so that the presence of a plant is known long before it is actually seen.

The accompanying plate illustrates a male plant.

DESCRIPTION:—Male plant about 30 cm. high. *Root-stock* thick, irregularly lobed, verrucose; stem short erect; leaves reduced to oblong obtuse or subacute scales up to 2 cm. long and 1·2 cm. wide. *Inflorescence* much-branched; flowers usually in pairs on short pedicels which are connate below. *Perianth-segments* navicular, almost patent, very thick and fleshy, subacute, 4 mm. long, 2·5 mm. wide; filaments 3 to nearly 4 mm. long, cylindrical; anthers terminal, scarcely wider than the filaments. Female plants very similar to the male, but rather shorter, flowers numerous in subglobose shortly stalked heads about 6 mm. in diameter. *Ovary* 1-3-celled;

ovule solitary, pendulous; stigma discoid, sessile; fruit a syncarpium; seed about 1 mm. long (*Flora Capensis*).

PLATE 176.—Fig. 1, scale-like leaf; Fig. 2, portion of male inflorescence; Fig. 3, single male flower, showing the three perianth-segments and the 3 stamens; Fig. 4, single stamen.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 177.

PROTEA PITYPHYLLA.

Cape Province.

PROTEACEAE. Tribe PROTEAE.

PROTEA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 169.

Protea pityphylla, *Phillips in Kew Bulletin*, 1910, 234; *Fl. Cap.* vol. v. sect. i. p. 594.

We have pleasure in showing for the first time this species of *Protea*, hitherto not figured in botanical publications. On a previous occasion we published an illustration of a broad-leaved variety, *P. pityphylla*, var. *latifolia* (see Plate 108). Another variety with short leaves resembling those of *P. rosacea* has been collected in the Ronde Bokkeveld, but it may be at once distinguished from this species by the foliaceous appendages of the outer bracts.

The late Dr. MacOwan distributed the species as a “sp. nov.” in 1888, and it is only within recent years that it has been at all largely collected.

The species has been successfully cultivated at the National Botanic Gardens, Kirstenbosch, and is well worth the attention of gardeners interested in plants typically South African.

Our plate was prepared from specimens presented by Dr. R. Marloth.

DESCRIPTION:—*Branches* glabrous; leaves 6 to 8 cm. long, about 1 mm. wide, needle-shaped, acute, pungent, channelled and prominently costate on the upper face, glabrous. *Head*, sessile, 4 to 4·5 cm. long, about 6·5 cm. in diameter, cernuous. *Involucral-bracts* 7-seriate, glabrous; outer ovate, acuminate, obtuse or acute, the lowest produced into long foliaceous appendages resembling the leaves, inner oblong, slightly concave, exceeding the flowers. *Perianth-sheath* 1·5 cm. long, dilated, 3-keeled and 7-nerved below, scarious, rufously setulose within in the upper part, otherwise glabrous; lip 5 mm. long, 3-toothed, 3-keeled, setose below; teeth subequal,

0·5 mm. long. *Stamens* all fertile; filaments 0·5 mm. long, dilated, concave; anthers oblong-linear, 3 mm. long, apical glands 0·25 mm. long, ovate, subacute, somewhat swollen on the inner face. *Ovary* 2 mm. long, obovate-oblong, covered with long reddish-yellow hairs; hypogynous scales 1 mm. long, oval-oblong; style up to 2·2 cm. long, widened and much compressed from the base upwards for 6 mm., then much constricted and strongly bent and subulate, the slender portion obliquely arching inwards, glabrous: stigma 3 mm. long, obtuse (*Flora Capensis*).

PLATE 177.—Fig. 1, single flower; Fig. 2, single flower opened; Fig. 3, lip of perianth-segments showing the three stamens; Fig. 4, receptacle.

F.P.S.A., 1925.



S. Gower del.

PLATE 178.

ALOE RUPESTRIS.

Namaqualand.

LILIACEAE. Tribe ALOINEAE.

ALOE, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 776.

Aloe rupestris, *Baker in Flora Capensis*, vol. vi. p. 326.

This remarkable *Aloe* belongs to the same section of the genus as *A. ferox*, which we figured on Plate 169, but differs in having leaves which are not prickly on the under surface. It is only found, so far as we know, in Namaqualand, where it grows on rocky ground, and was first distributed by MacOwan. The locality from which the original specimens came was between Port Nolloth and Spektakel. The plant grows to a height of 10 to 12 feet, and produces a large terminal panicle of racemes which overtops the leaves. The young flowers are greenish-yellow in colour, but become red when adult. This difference in colour between the young and adult flowers is not an uncommon character in the genus, and we have noted it before in *A. Wickensii* (Plate 41).

The plant from which our plate was prepared was presented by Dr. R. Marloth, and flowered at the Division of Botany, Pretoria, in September 1924.

DESCRIPTION:—*Stem* almost 2 m. high, thick. *Leaves* in a rosette at the apex of the stem, up to 60 cm. long, 5 cm. broad near the base, lanceolate-ovate, acuminate, not prickly on either side, with small deltoid marginal teeth. *Inflorescence* much-branched; racemes dense, 15 to 20 cm. long. *Bracts* small. *Pedicels* short. *Perianth* over 2 cm. long; segments divided almost to the base, oblong, with a distinct green keel. *Stamens* and *style* much exserted.

PLATE 178.—Fig. 1, plant, much reduced;
Fig. 2, median longitudinal section of flower;

Fig. 3, flowers in various stages of development; Fig. 4, stamen; Fig. 5, apex of style.

F.P.S.A., 1925.



S. Gower del.

PLATE 179.

SENECIO FULGENS.

Natal, Transvaal.

COMPOSITAE. Tribe SENECONIDEAE.

SENECIO, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. ii. p. 446.

Senecio fulgens, *Nicholson Dict. Gard.* vol. iii. p. 420; *Kleinia fulgens*, *Hook. Bot. Mag.* t. 5590.

This species of *Senecio* belongs to the section *Kelinoidei* of the genus, to which section *Senecio stapelliaeformis*, figured on Plate 28, is also referred. The species was first introduced into England from Natal by a Mr. Plant in 1886, and flowered at Kew the same year. From these specimens the plate in the "Botanical Magazine" was made.

A comparison of the present plate with that of the "Botanical Magazine" quoted above will show that our plant has a more lax habit, and this is to be explained by the fact that it is growing under the shelter of a large tree. The shape and dentition of the leaves vary considerably. The young leaves are narrowly (1·5 cm.) lanceolate, while adult leaves are broadly (4·5 cm.) obovate. Some of the leaves are quite entire, while others are remotely toothed. All these variations are found on the same plant. On young branches the leaves are grouped in more or less of a rosette at the apex of the branch.

The species is well worthy of cultivation, as it grows luxuriantly and flowers profusely, and often produces flowering stems over 30 cm. long. The flowers are coral-red (R.C.S., Plate XIII).

DESCRIPTION:—A herbaceous shrub up to ·6 m. high. *Leaves* more or less crowded at the base, 6 to 12 cm. long, 1·5 to 4·5 cm. broad, lanceolate, lanceolate-obovate to obovate, narrowed at the base, entire or remotely toothed, glaucous, fleshy; the older leaves channelled on the upper surface in the lower portion and distinctly keeled beneath. *Flowering stems* up to 30 cm. long, with scattered leaves 2 to 9 cm. apart, which decrease in size

upwards, usually simple. *Heads* homogamous, solitary. *Involucral-bracts* 1·9 cm. long, concrete, forming a tube 1 cm. in diameter and oblong in outline, produced into nine lanceolate lobes above. *Receptacle* slightly concave. *Corolla-tube* 1·9 cm. long, cylindric, very gradually widening upwards; lobes 2 mm. long, spreading, ovate-oblong, obtuse. *Anthers* blunt at the base, with a lanceolate apical appendage. *Style-branches* much recurved, obtuse, with marginal papillae. *Ovary* 5 mm. long, cylindric, glabrous. *Pappus* 1·2 cm. long, of many fine bristles. (National Herb., Pretoria, No. 2738.)

PLATE 179.—Fig. 1, plant, much reduced; Fig. 2, cross-section of leaf; Fig. 3, longitudinal section of head; Fig. 4, median longitudinal section of flower; Fig. 5, anthers; Fig. 6, style; Fig. 7, fruit and pappus.

F.P.S.A., 1925.

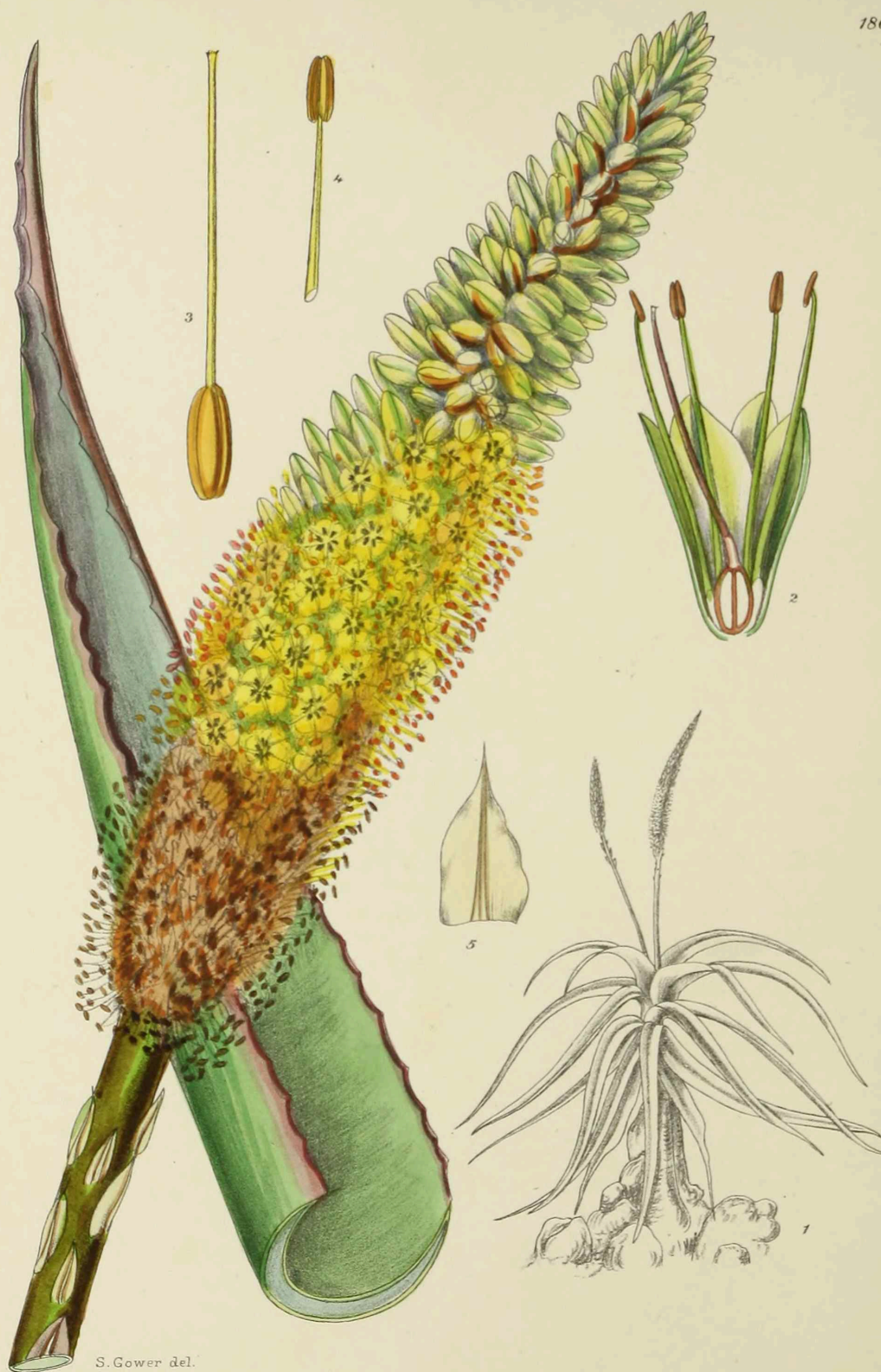


PLATE 180.

ALOE SESSILIFLORA.

Transvaal.

LILIACEAE. Tribe ALOINEAE.

ALOE, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 776.

Aloe sessiliflora, *Pole Evans in Trans. Roy. Soc. S. Afr.* vol. v. p. 708.

The species of *Aloe* figured on the accompanying plate belongs to the less conspicuous members of the genus. The small cream-yellow flowers do not make the inflorescence very attractive. *Aloe sessiliflora* was collected in the Barberton District by Mr. J. E. Wickens, and flowered at the Union Buildings for the first time in June and July of 1914. Mr. Geo. Thorncroft subsequently collected the plant near Barberton. During the winter months the leaves are of a distinct reddish colour, while in summer they are bright green. The flowers contain drops of very dark nectar at the base of the perianth.

Our plate was made from specimens growing at the Division of Botany, Pretoria.

DESCRIPTION:—*Stem* up to 90 cm. high. *Leaves* in a dense rosette at the apex of the stem, 45 to 60 cm. long. 6 to 8 cm. broad, spreading or recurved, fleshy, channelled above, convex beneath, with the margins toothed. *Peduncle* simple, 60 to 75 cm. long, laterally compressed, covered with numerous oblong brown bracts. *Spike* densely many-flowered, more or less cylindric; bracts 10 mm. long, 7 mm. wide, ovate-cuspidate, 3-nerved. *Flowers* campanulate-cylindric; perianth 14 mm. long; segments free; the outer 5 mm. broad, spatulate, fleshy-coloured with three longitudinal dark nerves; the inner 8 mm. broad, yellowish at the edges and with a reddish or greenish median line. *Stamens* and *style* protruding 8 to 10 mm. beyond the perianth. *Capsule* 9 to 10 mm. long, cylindric-oblong, enclosed in the dry peri-anth. *Seeds* 3 mm. long, three-angled, greyish, very narrowly winged. (National Herb., Pretoria, No. 2880.)

PLATE 180.—Fig. 1, habit; Fig. 2, median section of flower; Fig. 3, pistil; Fig. 4, stamen; Fig. 5, bract.

F.P.S.A., 1925.



PLATE 181.

STAPELIA GIGANTEA var. PALLIDA.

Transvaal (?).

ASCLEPIADACEAE. Tribe STAPELIEAE.

STAPELIA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. ii. p. 784.

Stapelia gigantea, *N.E. Br.* var. **Pallida**, *Phillips* var. nov., a typo
corolla depressiore differt.

The *Stapelia* figured on the accompanying Plate is very near *S. gigantea*, *N.E. Br.*, from which it differs, however, in having the disc portion of the corolla more depressed and being much lighter in colour. It appears to be an intermediate form between this species and *S. nobilis*, *N.E. Br.*, differing from the latter species in not having such a cup-shaped corolla-disc and not having the lobes of the outer corona 3-toothed. The locality in which the plant is found is not known, and the specimen from which our Plate was prepared was kindly presented by Dr. A. J. T. Janse, who grew it in his rockery.

Description:—*Stems* robust, bright green, finely pubescent, prominently ridged, with each ridge ending in a tooth-like leaf. *Flowers* solitary. *Pedice*l 5 cm. long, terete, finely pubescent. *Sepals* 1·1 cm. long, lanceolate, acute, pubescent. *Corolla* 25 cm. in diameter when expanded; lobes 10 cm. long, 3 cm. broad at the base, ovate, long-attenuate, pubescent without, transversely rugose and with fine purple hairs on the inner face; disc shallowly depressed, covered with long soft purple hairs. *Outer* corona-lobes oblong; inner corona-lobes deeply cleft, with the outer lobes plate-like and the inner lobes linear. (National Herb., Pretoria, No. 2891.)

PLATE 181.—Fig. 1, side view of flower;
Fig. 2, corona; Fig. 3, cross-section through

stem.

F.P.S.A., 1925.



F. Sewell del.

PLATE 182.

GLADIOLUS CRUENTUS.

Natal, Basutoland.

IRIDACEAE. Tribe IXIEAE.

GLADIOLUS, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 709.

Gladiolus cruentus, *Moore in Gard. Chron.* 1868, 1138; *Bot. Mag.* t. 5810;
Fl. Cap. vol. vi. p. 157.

This is the first opportunity we have had of figuring a species of *Gladiolus* belonging to the section *Cardinales*. As far as our records go the species appears to be little known to botanical science. In 1868 a Mr. Bull flowered it in his nursery at Chelsea, and it was from specimens supplied by Mr. Bull that the Plate in the *Botanical Magazine* was prepared. We are indebted to Mr. L. F. Wachter, who sent us specimens from Basutoland in 1923, for information about this beautiful plant. Mr. Wachter states that the plant grows in many places in the mountain area of Basutoland, and it is probable that the species is confined to the high mountain regions of Natal and Basutoland. While we have no definite information as to the precise habitat, there appears to be little doubt that it favours similar localities to *G. cardinalis* of the Cape Province, which is usually found growing near waterfalls.

DESCRIPTION:—*Corm* 3·5 cm. in diameter, with thick cylindric roots. *Leaves* about four, 16 to 30 cm. long, 1·5 to 2·5 cm. broad, ensiform, glabrous. *Spike* few-flowered. *Spathe-valves* large, lanceolate; the lower from 7 to 15 cm. long. *Perianth-tube* funnel-shaped, curved; upper segments 5 to 6·5 cm. long, obovate-spathulate; lower segments about 4 cm. long, with a white blotch at the throat covered with red spots. *Style-branches* papillose on the margins.

PLATE 182.—Fig. 1, median longitudinal section of flower; Fig. 2, style showing the style-branches.

F.P.S.A., 1925.



P. Badenhorst del.

PLATE 183.

GAZANIA SUFFRUTICOSA.

S.W. Africa.

COMPOSITAE. Tribe ARCTOTIDEAE.

GAZANIA, *Gaertn.; Benth. et Hook. f. Gen. Plant*, vol. ii. p. 459.

Gazania suffruticosa, *Muschler in Engl. Bot. Jahrb.* vol. 46, p. 120.

In a collection of succulents received from South-West Africa and planted at the Division of Botany, Pretoria, fruits of this *Gazania* were evidently mixed with the soil, as a plant grew in the succulent bed. It is a somewhat remarkable species and differs considerably in habit from the two species previously figured on Plates 51 and 64. The whole plant is covered with cobwebby hairs. The leaves are somewhat succulent, extremely brittle, and snap when bent; they are also easily snapped from the branches.

The single specimen in the garden has made vigorous growth and promises to do well under cultivation. It flowered in May of this year.

DESCRIPTION:—A branched herb up to 30 cm. high. *Branches* cobwebby. *Leaves* 5·5 to 7·5 cm. long, 1 to 1·5 cm. broad above, obovate-spathulate, narrowed into a petiole, pungent at the apex, irregularly lobed, or toothed, with the lobes and teeth tipped with a short pungent mucro, cobwebby above and beneath. *Flower-heads* in the axils of the leaves. *Peduncles* slightly longer, as long as or shorter than the leaves with tufts of cobwebby hairs. *Involucre* 1·8 cm. long, 6 mm. in diameter below, the concrete portion oblong in outline, inflexed at the base, sparsely cobwebby; outermost lobes 2·5 to 4 mm. long, ovate, acute; inner lobes 1 cm. long, ovate, acuminate, acute, with membranous margin. *Ray-florets* neuter; tube 9 mm. long, somewhat compressed; limb 1·6 cm. long, 7 mm. broad, obovate-elliptic, yellow, with a black eye-spot. *Disc-florets*: tube 7 mm. long, cylindric; lobes 1·5 mm. long, oblong, shortly acuminate, subobtuse. *Anthers* minutely

sagittate at the base. *Ovary* very villous; style-branches linear, obtuse. *Pappus* of many delicate very narrow scales 5 mm. long.

PLATE 183.—Fig. 1, involucre; Fig. 2, longitudinal section through involucre; Fig. 3, ray-floret; Fig. 4, disc-floret.

F.P.S.A., 1925.

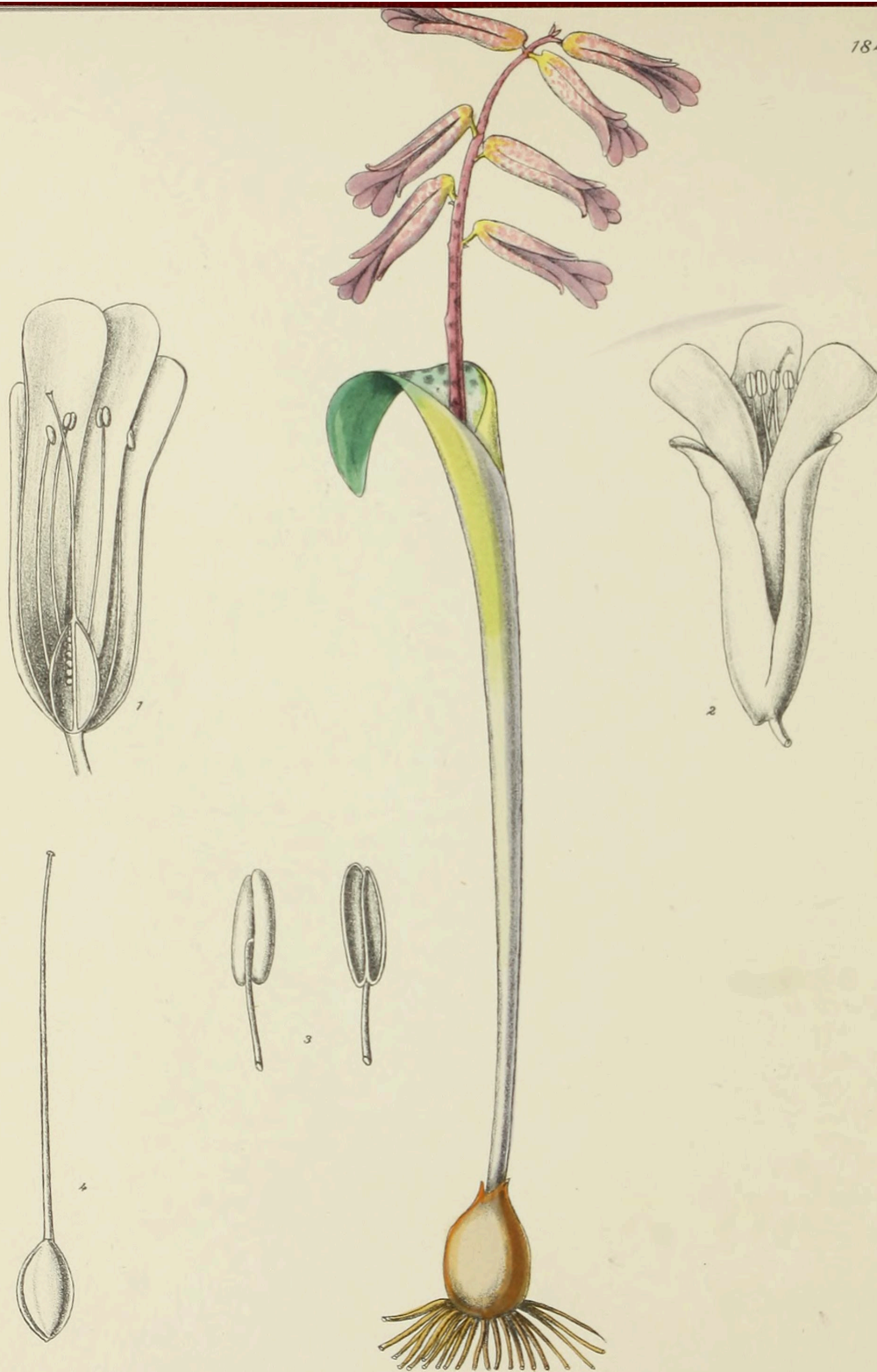


PLATE 184.

LACHENALIA RUBIDA var. TIGRINA.

Cape Province.

LILIACEAE. Tribe SCILLEAE.

LACHENALIA, *Jacq.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 807.

Lachenalia rubida, *Jacq.* var. **tigrina**, *Bkr. Fl. Cap.* vol. vi. p. 424.

On Plate 158 we figured a species *Lachenalia pendula* belonging to the same subgenus (*Eulachenalia*) as the species here described. *Lachenalia rubida* is a graceful little plant, though not so handsome as some species of the genus. It was figured by Jacquin between the years 1786-1793, from plants cultivated in Europe, so that it has been known for over 130 years to botanists.

The species appears to be confined to the Clanwilliam, Calvinia and van Rhynsdorp Districts of the Cape Province, and as far as our records go does not appear to have been frequently collected.

We are indebted to Mrs. E. Rood of van Rhynsdorp for the specimens from which our Plate was prepared.

DESCRIPTION:—*Bulb* 1·5 cm. in diameter at the base, ovoid, white. *Produced leaves* 1 or 2, as long or slightly longer than the naked portion of the peduncle, with the free portion 1·8 to 2·2 cm. broad, elliptic-oblong or ovate-oblong, obtuse, with dark brown blotches on a dark green background, sometimes the spots absent, convolute and clasping the peduncle for the greater portion of its length. *Peduncle* with maroon-coloured spots on a yellowish background; in concolorous leaves peduncle almost uniformly coloured. *Inflorescence* 6-7-flowered. *Bracts* forming small pockets from which the flowers arise. *Pedicels* 3 mm. long. *Outer perianth-segments* 0·5 cm. shorter than the inner, slightly gibbous at the base, thickly speckled with

red spots on a yellowish background; inner perianth-segments oblong, obtuse. *Stigma* capitate. (National Herb., Pretoria, No. 2901.)

PLATE 184.—Fig. 1, median longitudinal section of a flower; Fig. 2, single flower; Fig. 3, anthers; Fig. 4, pistil.

F.P.S.A., 1925.



P. Badenhorst del.

PLATE 185.
PROTEA MUNDII.

Cape Province.

PROTEACEAE. Tribe PROTEAE.

Protea, Linn.; Benth. et Hook. f. *Gen. Plant.* vol. iii. p. 169.

Protea Mundii, Klotz in Otto and Dietr. *Garten-Zeit.* 1838, 113; *Fl. Cap.*
vol. v. sect. i. p. 579.

The species of *Protea* figured on the accompanying Plate represents a member of the section *Exertae* of the genus. This section comprises four species which may be readily recognised by the fact that in the mature head the perianth is spirally coiled in the open flowers, leaving the style exerted. *Protea Mundii* is closely related to another species found in the same localities, viz. *P. laticolor*, Salisb., and the two were confused by E. Meyer, who named them both *P. penicillata*. As will be seen from the illustration, the stigma is very distinctly capitate at the apex, and as this character is unique in the genus the species can be easily recognised.

The plant is found in the mountains round Worcester, extends into the George, Knysna and Humansdorp Districts, and then passes through Uitenhage and Stutterheim into the Transkei.

The specimens were collected by Dr. I. B. Pole Evans, C.M.G., in the George District in July 1925.

DESCRIPTION:—Branches tomentellous to tomentose above. *Leaves* 4 to 11 cm. long, 1·2 to 3·2 cm. broad, lanceolate or lanceolate-elliptic, subobtusely, narrowing at the base, distinctly veined, glabrous or the youngest leaves sometimes loosely pilose. *Head* sessile, 7 to 8 cm. long, about 5 cm. in diam. *Involucral-bracts* 11-12-seriate; outer ovate, obtuse, silky on the back, green, ciliate; inner oblong or spathulate-oblong, whitish pubescent to tomentose,

fringed with white cilia, shorter than the styles; perianth-sheath 4 cm. long, slender and thin above the middle, gradually dilated and 5-nerved below, not keeled, the upper half at length coiled up, loosely hairy; lip 1·5 cm. long, tridentate, glabrous, with a dense tuft of hairs at the apex; lateral teeth 2 mm. long; median tooth 1·5 mm. long; stamens all fertile; filaments 1 mm. long, channelled down the middle; anthers linear, 6 mm. long; apical glands 0·5 mm. long, ovate, subacuminate, subacute, keeled on the inner face; ovary covered with a tuft of long brown hairs; style 5 cm. long, almost straight, keeled on one side, compressed above the ovary, then more or less terete, glabrous; stigma 6 mm. long, furrowed, subcapitate at the apex, abruptly and obliquely passing into the much stouter style. (National Herb., Pretoria, No. 2918.)

PLATE 185.—Fig. 1, single flower; Fig. 2, receptacle.

F.P.S.A., 1925.



P. Badenhorst del.

PLATE 186.

STRELITZIA REGINAE.

Cape Province.

SCITAMINEAE. Tribe MUSEAE.

STRELITZIA, *Ait. in Benth. et Hook. f. Gen. Plant.* vol. iii. p. 656.

Strelitzia Reginae, *Banks in Ait. Hort. Kew.* ed. 1, i. 285, t. 2; *Fl. Cap.*
vol. v. sect. iii. p. 316.

In 1792 this plant was figured in the *Botanical Magazine*, (Plate 119), but had previously been figured by Sir Joseph Banks. *Strelitzia Reginae*, which was introduced into the Royal Gardens at Kew in 1773, excited a considerable amount of interest when it flowered. In the number of the *Botanical Magazine* quoted above a double Plate was devoted to the illustration of the flowers so as to give “readers an opportunity of seeing a coloured representation of one of the most scarce and magnificent plants introduced into this country.” That the plant is a particularly handsome one there can be no doubt and it has been appropriately named the “Bird-of-paradise flower”; the Afrikaans name is “Gele piesang.” The species is native of the south-eastern and eastern districts of the Cape Province.

The specimen illustrated on the accompanying Plate was grown at the Division of Botany, Pretoria.

The following description is taken mainly from the *Flora Capensis*.

DESCRIPTION:—Stemless; leaves oblong-lanceolate, acute, cuneate at the base, up to 0·5 m. long and 10 cm. wide, entire, undulate, especially in the lower part, quite glabrous, bright green above, glaucescent beneath; petiole up to 1·25 m. long; peduncle as long as the petiole. *Bracts* tubular, oblique and acute at the mouth, uppermost one cymbiform, acuminate, up to 20 cm. long, green, edged with purple. *Sepals* lanceolate, 7 to 10 cm. long, orange-

yellow. *Petals* dark blue, blade of the lower two 5 cm. long, with a rounded basal auricle; claw 2·5 cm. long; upper petal ovate, 2·5 cm. long. *Stamens* reaching to the top of the longer petals; anthers narrowly linear, twice as long as the filaments; style exserted, with 3 linear branches 2·5 cm. long. *Fruit* a capsule; seeds covered with reddish woolly hairs. (National Herb., Pretoria, No. 2915.)

PLATE 186.—Fig. 1, plant much reduced;
Fig. 2, median longitudinal section of a
flower.

F.P.S.A., 1925.



PLATE 187.

ALOE ARBORESCENS var. FRUTESCENS.

Transvaal.

LILIACEAE. Tribe ALOINEAE.

ALOE, *Linn.; Benth. et Hook. f. Gen. Plant*, vol. iii. p. 776.

Aloe arborescens, *Miller, Gard. Chron.* ed. viii. No. 3; var. ***frutescens***,
Bkr. Fl. Cap. vol. vi. p. 322.

This *Aloe* is widely distributed along the eastern mountain range and is known to extend northwards as far as the Victoria Falls. The many stems which grow together give the plant a bush-like appearance which makes a very effective display in a large rockery, and it has the advantage over many other species of the genus in that the inflorescence is not nipped by a severe frost. The species also lends itself admirably to growing on rough stone pillars forming an entrance to a drive.

Our specimen was collected on the hills near Haenertsburg on the Drakensbergen in June 1914, and cultivated at the Division of Botany, Pretoria.

DESCRIPTION:—An arborescent branched shrub. *Leaves* up to 0·5 m. long, about 4 cm. broad at the base, gradually tapering to the apex, almost flat on the upper surface, convex on the lower surface, with ovate somewhat incurved teeth 1 cm. apart below, about 1·3 cm. apart above, glabrous. *Inflorescence* solitary or 2-3 from each rosette of leaves, unbranched. *Peduncle* 7 mm. in diameter, cylindric with scattered membranous bracts. *Floral-bracts* membranous, 1·3 cm. long, 9 mm. broad, oblong, with straight margins, distinctly veined. *Pedicels* 2 cm. long, elongating in the old flowers. *Raceme* dense, up to 22 cm. long. *Young flowers* erect-spreading, cylindric, with a subacute apex; old flowers pendulous. *Perianth* (in open flowers) 3·2 cm. long; outer segments reddish, 5 mm. broad, obtuse; inner

segments white, 9 mm. broad, with a distinct mid-rib, reddish below, greenish above, obtuse. *Stamens* projecting. *Ovary* 8 mm. long, cylindric; style 3·2 cm. long, cylindric; stigma minute. (National Herb., Pretoria, No. 2904.)

PLATE 187.—Fig. 1, young and adult flowers; Fig. 2, median longitudinal section of a flower; Fig. 3, pistil; Fig. 4, bract.

F.P.S.A., 1925.



PLATE 188.

DERMATOBOTRYS SAUNDERSII.

Cape Province, Natal.

SCROPHULARIACEAE.

DERMATOBOTRYS, *Bolus in Hook. Ic. Pl.* 1940.

Dermatobotrys Saundersii, *Bolus in Hook. Ic. Pl.* Zwilt's 1940; *Fl. Cap.*
vol. iv.
sect. ii. p. 206.

This interesting member of the *Scrophulariaceae* is peculiar in that it grows as an epiphyte on other trees. Miss Pegler records it from the Kentani District as growing on the Cape Chestnut (*Calodendron capense*). It ranges from Kentani to Zululand. It differs also from most members of the family in having 5 well-developed stamens, and for this reason was first referred to the family *Solanaceae*, but the straight or almost straight embryo indicates an affinity with the family *Scrophulariaceae*.

Mr. Saunders, who first collected the plant, described it as a parasite which killed the host on which it grew, but the late Dr. Medley Wood states that the plant has a tendency to fix itself to trees already dead.

We are indebted to Mr. K. B. Jameson, Scottsville, Maritzburg, for the specimen from which our illustration was prepared.

DESCRIPTION:—A glabrous epiphytic shrub; root-stock 1·25 m. high, about 1 cm. thick but increasing towards the top to 5 cm., furrowed transversely as in a *Dahlia* root; rootlets fibrous; stems more or less quadrangular; ultimate branchlets 1·5 to 3 mm. thick. *Leaves* opposite, decussate, ovate or elliptical, acute or broadly pointed at the apex, more or less narrowed at the entire base, strongly toothed or repand-dentate, fleshy, red-veined, turning black-green in the dried state, 5 to 15 cm. long, 2·5 to 9 cm. broad; petioles 1 to 5

cm. long. *Flowers* clustered at the nodes on the branchlets, usually three together, bracteate at the base, about 4 cm. long; peduncles 1 to 3 mm. long, spreading; bract elliptic-linear, acute at both ends, about 2 cm. long, 5 mm. broad; calyx-segments lanceolate, acute, glabrous, 3 to 5 mm. long. *Corolla* red; tube beset inside towards the base with stiff broad white hairs; lobes about 5 mm. long; anthers glabrous; style glabrous, slender, tapering towards the stigma. *Ovary* ovoid-conical, glabrous; ripe berry ovoid, blunt, smooth, about 2 cm. long, 1·8 cm. broad, green; embryo about 1/3 to 3/4 of the seed in length. (National Herb., Pretoria, No. 2917.)

PLATE 188.—Fig. 1, single flower; Fig. 2, median longitudinal section of flower; Fig. 3, pistil.

F.P.S.A., 1925.

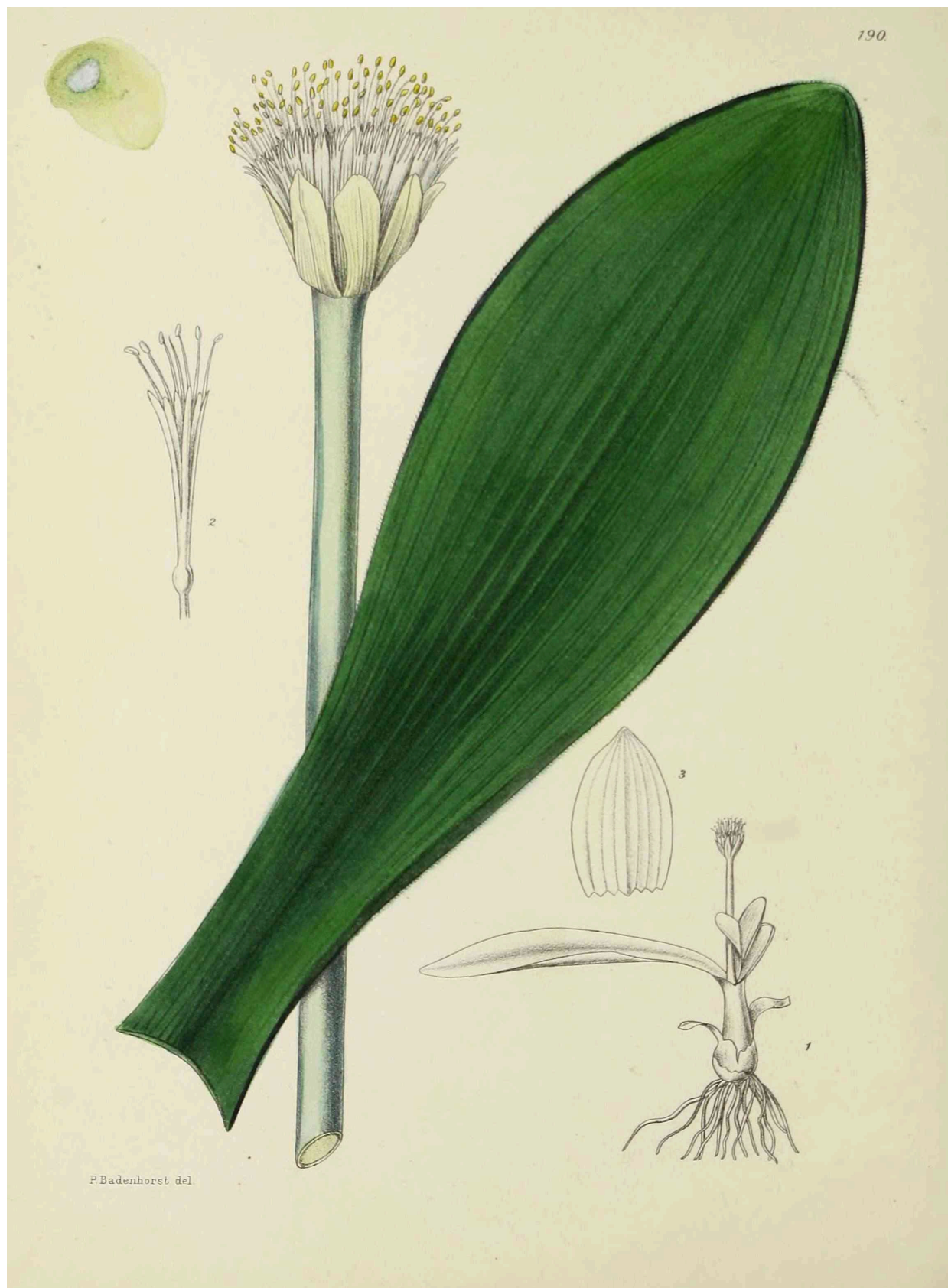


PLATE 189.

CRASSULA LATICEPHALA.

Cape Province.

CRASSULACEAE.

CRASSULA, *Linn.; Benth. et Hook. f. Gen. Plant*, vol. i. p. 657.

Crassula laticephala, *Schonl. in Rec. Albany Mus.* vol. ii. p. 457.

The dwarf *Crassula* figured on the accompanying Plate was described by Dr. S. Schonland in 1913 from a specimen collected in the Riversdale District by Mr. J. E. A. Volschenk. The species is closely allied to *C. congesta*, N.E.Br., which we figured on Plate 115, and Dr. Schonland suggests that when more material is available it might be considered a variety of the latter species.

Crassula laticephala is confined, so far as we know, to the Riversdale District, and there only found in the Klein Karroo. We are indebted to Dr. J. Muir of Riversdale for the specimen figured.

DESCRIPTION:—A dwarf succulent 4·5 cm. high. *Leaves* fleshy, 4-ranked, strongly reflexed, 3 cm. long, 1·2 cm. broad at the base, ovate, acuminate, obtuse, flat above with a distinct keel, somewhat convex beneath, scurfy. *Flowers* in heads 2·5 to 3 cm. in diameter surrounded by floral-leaves. *Floral-leaves* 1·2 cm. long, 1·5 cm. broad at the base, ovate, suddenly contracted into a fleshy cylindric appendage. *Receptacle* convex. *Floral-bracts* 4·5 mm. long, linear, ciliated. *Calyx-lobes* 3 mm. long, linear, cucullate at the apex, ciliate. *Petals* 5·5 mm. long, linear, slightly broadened below, concave. *Filaments* 1 mm. long, slender; anthers 1·25 mm. long, oblong. *Carpels* 2 mm. long. *Hypogynous scales* transversely oblong, on distinct stalks. (National Herb., Pretoria, No. 2936.)

PLATE 189.—Fig. 1, flower-bud; Fig. 2, petal with 2 stamens; Fig. 3, carpels; Fig. 4, longitudinal section of head showing convex receptacle; Fig. 5, cross-section through leaf; Fig. 6, leaf.

F.P.S.A., 1925.



P. Badenhorst del.

PLATE 190.

HAEMANTHUS ALBIFLOS.

Cape Province.

AMARYLLIDACEAE. Tribe AMARYLLEAE.

HAEMANTHUS, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 730.

Haemanthus albiflos, *Jacq. Hort. Schoenbr.* i. 31, t. 59; *Fl. Cap.* vol. vi.
p. 235.

This species of *Haemanthus* was known to European horticulturists almost 150 years ago and was first shown in colour by Jacquin in 1797. In the *Botanical Magazine* it was again figured in 1810 (Plate 1239) from plants which flowered in the greenhouse of Messrs. Lee and Kennedy of Hammersmith. The plant has been recorded from the Uitenhage, Graaf Reinet and Somerset East Divisions of the Cape Province, but does not appear to have been extensively collected. In general habit *H. albiflos* resembles *H. natalensis* figured in Plate 32, but the leaves are thicker and more fleshy.

The species responds well to proper cultivation and has been successfully grown at the Division of Botany, Pretoria. It was from plants grown at Pretoria that our Plate was prepared.

DESCRIPTION:—*Bulb* tunicated, 4·5 cm. in diameter; tunics fleshy, white becoming green. *Leaves* 4, contemporary with the flowers, 42 to 46 cm. long, 10 cm. broad at the widest part, tongue-shaped, obtuse, narrowed to the base, with ciliated margins, dark green on the upper surface, paler on the lower surface. *Peduncle* 16 cm. long, 1·3 cm. in diameter, erect, compressed, glabrous. *Inflorescence* a dense umbel, 2·5 cm. in diameter. *Involucral-bracts* 5, ascending, white with 6 to 9 distinct green nerves, short mucronate, ciliated with reflexed hairs. *Pedicels* 4 to 5 mm. long, glabrous. Flowers white. *Perianth-tube* subcylindric, glabrous; segments 1·7 cm. long, linear,

obtuse. *Stamens* 6; filaments subulate; anthers versatile. *Ovary* globose, 3 mm. in diameter; style subulate; stigma minutely tricuspidate. (National Herb., Pretoria, No. 2933.)

PLATE 190.—Fig. 1, whole plant, much reduced; Fig. 2, single flower; Fig. 3, involucral bract.

F.P.S.A., 1925.



PLATE 191.

CEROPEGIA HAYGARTHII.

Natal, Cape Province.

ASCLEPIADACEAE. Tribe CEROPEGIEAE.

CEROPEGIA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. ii. p. 779.

Ceropegia Haygarthii, *Schltr. in Engl. Bot. Jahrb.* vol. xxxviii. 46, fig. 7 A;
Fl. Cap. vol. v. sect. i. p. 813.

This beautiful species of *Ceropegia* is almost unique in the genus on account of the peculiar formation of the corolla-lobes. The illustration should be compared with those given on Plates 39, 44 and 143, and the striking differences will then be seen. It is closely related to *C. tristis*, Hutch. described on Plate 44, but the corolla-lobes are produced into a much longer cylindric portion and the calyx-lobes are longer.

The specimen from which our illustration was made flowered in the greenhouse at the Division of Botany in July 1925, and like the other species of the genus always attracts attention owing to the peculiar shape of the flowers.

DESCRIPTION:—*Stem* climbing, fleshy, 3 to 4 mm. thick, glabrous. *Leaves* small, fleshy, flat, 0·6 to 3·5 cm. long, 0·3 to 2·5 cm. broad, ovate or ovate-lanceolate, acuminate. *Cymes* 1-2-flowered, lateral at the nodes. *Peduncles* 2 to 4 cm. long, glabrous. *Bracts* 2 to 4 mm. long, subulate. *Pedicels* 1 to 1·4 cm. long, glabrous. *Sepals* 1 cm. (0·5 cm., *Schlechter*) long, subulate, glabrous. *Corolla* abruptly bent at a right angle near the base. *Tube* (following the bend) about 3·5 cm. long, according to a drawing, or about 2·5 cm. long in dried flowers, globosely inflated at the base, cylindric above, enlarging (according to a drawing) to about 2 cm. in diam. at the mouth, pinkish-white or greenish tinted, spotted with violet, glabrous outside, pilose with very fine long hairs within. *Lobes* free at the base, abruptly inflexed

over the mouth of the tube and produced beneath into broad triangular partition-like green plates or keels, meeting at the centre and connate into a slender erect column 1 to 1·4 cm. long, then again becoming free and expanding into elliptic-lanceolate replicate segments connate at the tips, forming a small apical ellipsoid cage-like body 5 to 6 mm. long, ciliate on the margins, dull purple or purple-brown. *Corona* in the flowers seen much eaten by insects, but apparently the outer corona is cupular, with 5 acutely bifid lobes rising to the level of the top of the staminal column, ciliate and hairy within with long fine hairs. *Inner* corona-lobes 2 mm. long, linear or linear-spathulate, connivent-erect over the staminal column, with very revolute tips. (*Fl. Cap.*—National Herb., Pretoria, No. 2932.)

PLATE 191.—Fig. 1, calyx; Fig. 2, ground plan of corolla from above; Fig. 3, terminal portion of corolla lobes; Fig. 4, corona.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 192.

PROTEA HARMERI.

Cape Province.

PROTEACEAE. Tribe PROTEAE.

PROTEA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 169.

Protea Harmeri, *Phillips in Kew Bulletin* 1911, p. 83; *Fl. Cap.* vol. v. sect. i. p. 593.

A comparison of the accompanying Plate should be made with the illustration on Plate 108. Both the species figured belong to the same section of the genus *Protea*, all characterised by very narrow leaves.

Protea Harmeri was first found by Dr. (now Sir Sydney) Harmer on a hill near Matjesfontein when he visited South Africa in 1905; it was subsequently collected by Dr. R. Marloth on the Wittebergen and again later in the Zwartberg Pass between Oudtshoorn and Prince Albert. The specimen figured was collected near the summit of the Zwartberg Pass in July 1925. The plant is a bush about 6 ft. high and is extremely handsome, as the dark heads show up in strong relief from the greyish-green leaves.

DESCRIPTION:—A bush about 1 m. high. *Branches* greyish tomentellous above, becoming glabrous. *Leaves* 4·5 to 6·5 cm. long, 2·5 to 3·5 mm. broad, linear, obtuse to subacute with a callous point attenuated at the base, margins recurved; youngest leaves finely villous at the base. *Head* sessile, 2·5 cm. long, about 2·5 cm. in diam., globose; receptacle convex. *Involucral-bracts* 10-11-seriate; outer ovate, obtuse, glabrous or the lowest very finely pubescent, with membranous ciliate margins; inner oblong-spathulate, obtuse, brick-red, recurved above, glabrous or minutely pubescent, not equalling the styles. *Perianth-sheath* 1·5 cm. long, 0·75 mm. broad, dilated above, 3-keeled and 3-nerved below, glabrous or hirsute at the apex; lip 4 mm. long, 3-toothed, rufously setulose, glabrescent on the back;

teeth subequal, 0·25 mm. long, the middle one smaller. *Stamens* all fertile, subsessile; anthers linear, 3 mm. long; apical glands 0·25 mm. long, ovate, obtuse, swollen on the inner face. *Ovary* 2 mm. long, oblong-obovate in outline, covered with long brown hairs; hypogynous scales 1 mm. long, 0·25 to 0·5 mm. broad, oblong, obtuse; style 2 cm. long, falcate, arching over the centre of the head, terete above, flattened and hollow below, glabrous; stigma 2·25 mm. long, linear, obtuse, grooved, passing into the style. (*Fl. Cap.*—National Herb., Pretoria, No. 2916.)

PLATE 192.—Fig. 1, single flower; Fig. 2, posterior perianth lobe; Fig. 3, pistil; Fig. 4, receptacle.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 193.

SUTHERLANDIA FRUTESCENS.

Cape Province, Orange Free State, Natal, Transvaal.

LEGUMINOSAE. Tribe GALEGEAE.

SUTHERLANDIA, *R.Br.; Benth. et Hook. f. Gen. Plant.* vol. i. p. 503.

Sutherlandia frutescens, *R.Br. Hort. Kew.* ed. 2, p. 327; *Fl. Cap.* vol. ii.
p. 212.

This plant was named by the great English botanist in honour of James Sutherland, one of the earliest superintendents of the Edinburgh Botanic Gardens. The genus has only one species, which is widely distributed throughout South Africa, and is probably the most handsome native plant in the family *Leguminosae*. According to the *Botanical Magazine*, in which the plant was figured (Plate 181), *Sutherlandia frutescens* was known in cultivation as early as 1683, and later became generally known in European gardens.

Sutherlandia frutescens forms a small bush, but may grow to 3-4 ft. high, and is found usually in fairly dry places. It is known under several local names, but probably the most common is “Kanker bos” (cancer-bush), as it is reputed to be a cure for cancer.

As the plant is easily raised from seed, and the flowers are particularly beautiful, it should be more generally cultivated.

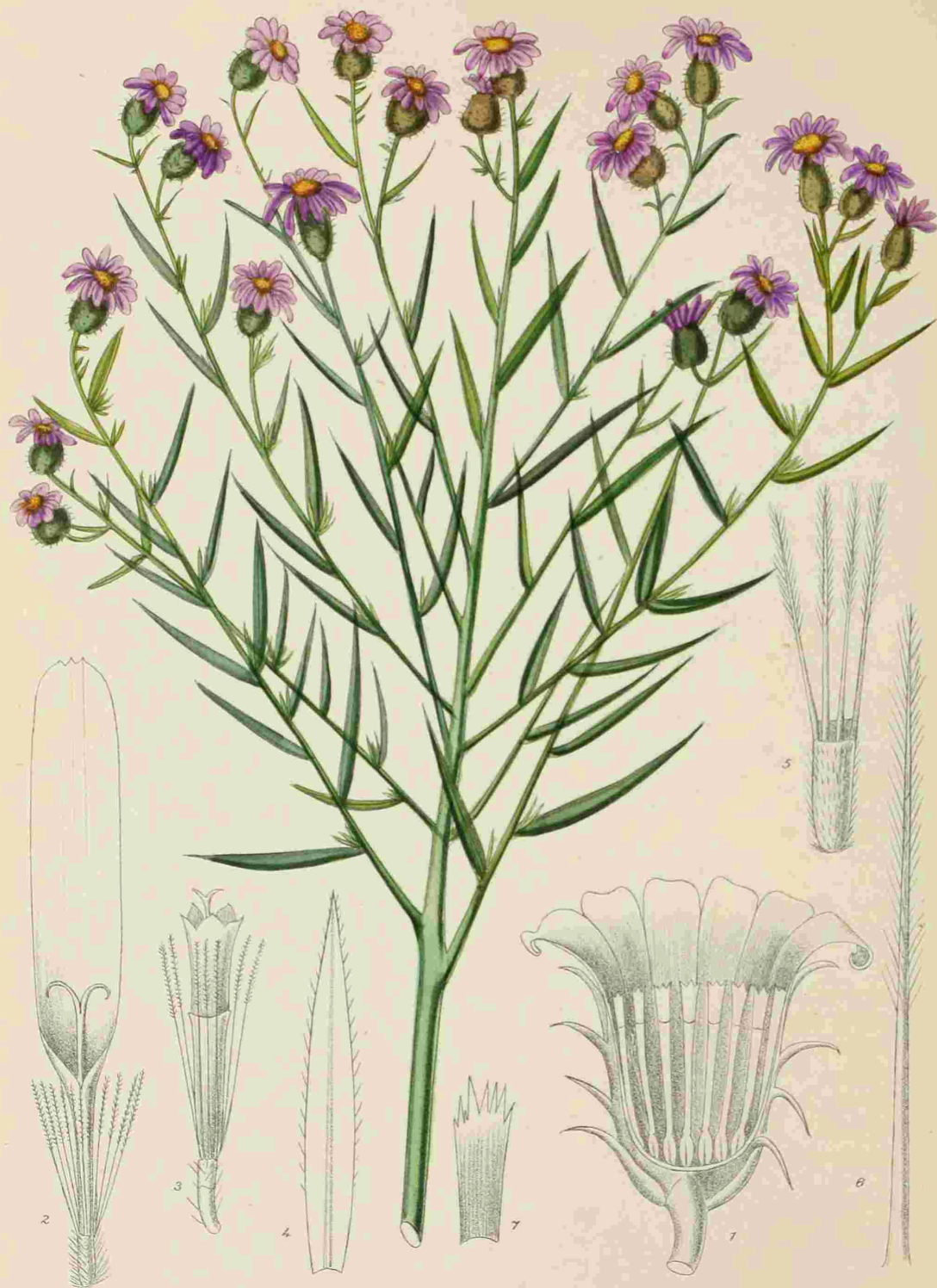
We are indebted to Mr. C. A. Smith, B.Sc., for the specimens which he collected at Fauresmith in the Orange Free State.

DESCRIPTION:—A shrub. *Branches* pubescent, at length becoming glabrous. *Leaves* 6 to 9 cm. long, imparipinnate; leaflets alternate or opposite, 1 to 2 cm. long, 3 to 6 mm. broad, lanceolate or oblong, obtuse, thinly pubescent. *Inflorescence* an axillary raceme, peduncle and pedicels

covered with short stiff sparse hairs. *Floral-bracts* 3 mm. long, oblong. *Pedicels* 1·2 cm. long. *Calyx-tube* 1 cm. long, deeply campanulate; lobes 4·5 mm. long, ovate, acuminate. *Vexillum* 3 cm. long, 1·2 cm. broad, obovate, acuminate; alae 8 mm. long, 1·5 mm. broad, oblong, obliquely clawed; keel 3·5 cm. long, 8 mm. broad, more or less oblong, with a linear claw 1·2 cm. long. *Stamens* diadelphous; filaments filiform. *Ovary* stalked, 1·5 cm. long, linear, with many ovules; style 1·5 cm. long, bearded on the inner face. *Pod* membranous, inflated. (National Herb., Pretoria, No. 2943.)

PLATE 193.—Fig. 1, median longitudinal section of flower.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 194.

ORNITHOGALUM ODORATISSIMUM.

Cape Province.

LILIACEAE. Tribe SCILLEAE.

ORNITHOGALUM, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 815.

Ornithogalum odoratissimum *C. A. Smith*, sp. nov. *Bulbus* ovoideo-globosus,

2 cm. diam., tunicis brunneis membranaceis. *Folia* 2-6, 6-10 cm. longa, anguste linearia, basi leviter dilatata, glabra. *Pedunculus* ad 14 cm. longus, cylindricus, glaber. *Inflorescentia* ad 6 cm. longa, pauci-vel multiflora. *Bracteae* 1-1·8 cm. longae, ovato-acuminatae, membranaceae. *Pedicelli* 0·5-1 cm. longi, teretes. *Segmenta perianthii* 1·5 cm. longa, oblonga, apice inflexa pubescentia obtusa. *Filamenta* 0·5-1 cm. longa, basi 2 mm. lata; antherae oblongae. *Ovarium* 5 mm. longum, sessile. *Stylus* 1·5 cm. longus, teres, apice minute capitatus, penicillatus. *Fructus* trigonus.

This Plate should be compared with that of *O. Roodeae* figured on Plate 75, as the two species are no doubt closely related though differing much in the colour of the flowers and the shape of the leaves. Like *O. Roodeae* the flowers are very sweet-scented, a character not usually met with in the genus, and a single inflorescence is sufficient to saturate a large room with a pleasant scent, and even when planted out in the garden the strong perfume is noticeable for a considerable distance from the plant. The plant has been compared with Jacquin's figure of *O. suaveolens*, from which it differs in being much smaller, and as it has been figured it was thought better to describe it than to definitely state that it is *O. suaveolens*.

We are indebted to Mrs. E. Rood of van Rhynsdorp for the specimens.

DESCRIPTION:—A simple erect herb. *Bulb* 2 cm. in diameter, 2·5 cm. high, ovoid, with brown tunics. *Leaves* 2 to 6, 6 to 10 cm. long, 2 to 3 cm. broad at the base, linear to linear-lanceolate, deeply channelled, glabrous. *Peduncle* up to 14 cm. long, terete, glabrous. *Raceme* few to many-flowered, up to 6 cm. long. *Bracts* 1 to 1·8 cm. long, ovate-acuminate, membranous, dry, partly sheathing the pedicels. *Pedicels* 0·5 to 1 cm. long, cylindric, erect-spreading. *Perianth-segments* 1·5 cm. long, oblong, inflexed obtuse and pubescent at the apex, white with a broad green 3-nerved band. *Stamens* slightly shorter than the perianth-segments; filaments 2 mm. broad at the base; anthers 2 mm. long, oblong, versatile. *Ovary* 5 mm. long, sessile; style subequalling the perianth-segments, sub-capitate and penicillate at the apex. *Fruit* (immature) deeply and bluntly 3-lobed. (National Herb., Pretoria, No. 2941.)

PLATE 194.—Fig. 1, median longitudinal section of flower; Fig. 2, cross-section of ovary.

F.P.S.A., 1925.



PLATE 195.

ATHRIXIA ELATA.

Basutoland, Cape Province, Natal, O.F.S., Transvaal.

COMPOSITAE. Tribe INULOIDEAE.

ATHRIXIA, *Ker.; Benth. et Hook. f. Gen. Plant*, vol. ii. p. 328.

Athrixia elata, *Sond. in Linn.* vol. xxiii. p. 67; *Fl. Cap.* vol. iii. p. 292.

The plant figured on the accompanying Plate is a much-branched shrub almost 1 m. high, and may be found in flower on the hills surrounding Pretoria during the month of August. It grows in masses in certain localities and flowers very profusely; if brought under cultivation it could no doubt be much improved so as to make it a desirable garden plant.

Athrixia elata is found on the Drakensberg in Natal, in Basutoland, the eastern Free State and on the spurs of the Drakensberg in the northern Transvaal, but extends westwards in the Transvaal as far as Rustenburg. The travellers Burke and Zeyher collected specimens almost 100 years ago on the Magaliesberg, and Cooper also came across the plant in his journey through Basutoland in 1861.

We are indebted to Mr. C. A. Smith, B.Sc., for the specimens from which our Plate was prepared.

DESCRIPTION:—A much-branched shrub. *Leaves* alternate, sessile, 0·5 to 2·5 cm. long, linear, acute, pungent, with revolute margins, glabrous above, woolly beneath. *Heads* subsessile, solitary, terminal, 2·5 to 3 cm. in diameter (including the rays). *Involucral-bracts* in many rows, unequal; the outer gradually shorter, recurved, aristate, 3 to 4 mm. long, 1 mm. broad, narrowly ovate with the apices edged with brown to black, woolly; inner 5 to 6 mm. long, 1 mm. broad, linear-oblong; innermost 7 to 8·5 mm. long, 0·5 to 1 mm. broad, narrow-linear to oblong-elliptic, with membranous margins.

Receptacle flat, nude, honeycombed with the margins of the cells evident. *Ray-florets* female; tube 4 to 5 mm. long, cylindric, widening above, glabrous; lamina oblong-elliptic. *Ovary* terete, pilose; style cylindric, glabrous, deeply 2-cleft. *Disc-florets* hermaphrodite, fertile. *Corolla* 6 mm. long, tubular, cylindric below, then articulated and the uppermost one-third slightly wider, glabrous. *Anthers* tailed at the base, with an ovate obtuse apical appendage. *Ovary* pilose; style cylindric, glabrous, deeply 2-cleft; branches 1 mm. long, linear, truncate, penicillate. *Pappus-bristles* equalling the corolla-tube, rigid, persistent, subplumose in the uppermost one-third, alternating with minute serrulate scales. (National Herb., Pretoria, 2942.)

PLATE 195.—Fig. 1, diagrammatic longitudinal section of head; Fig. 2, ray-floret; Fig. 3, disc-floret; Fig. 4, involucre-bract; Fig. 5, pappus enlarged; Fig. 6, pappus-bristle; Fig. 7, fimbriated pappus-scale.

F.P.S.A., 1925.

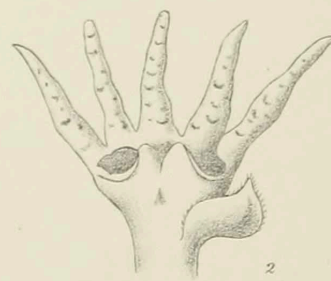
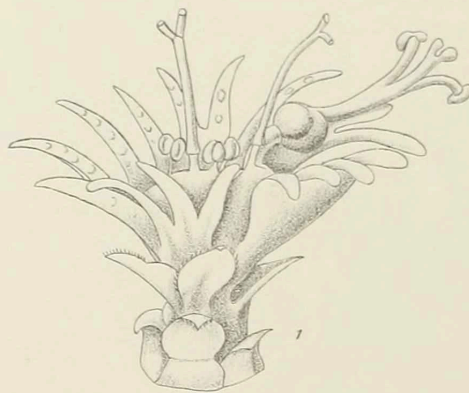
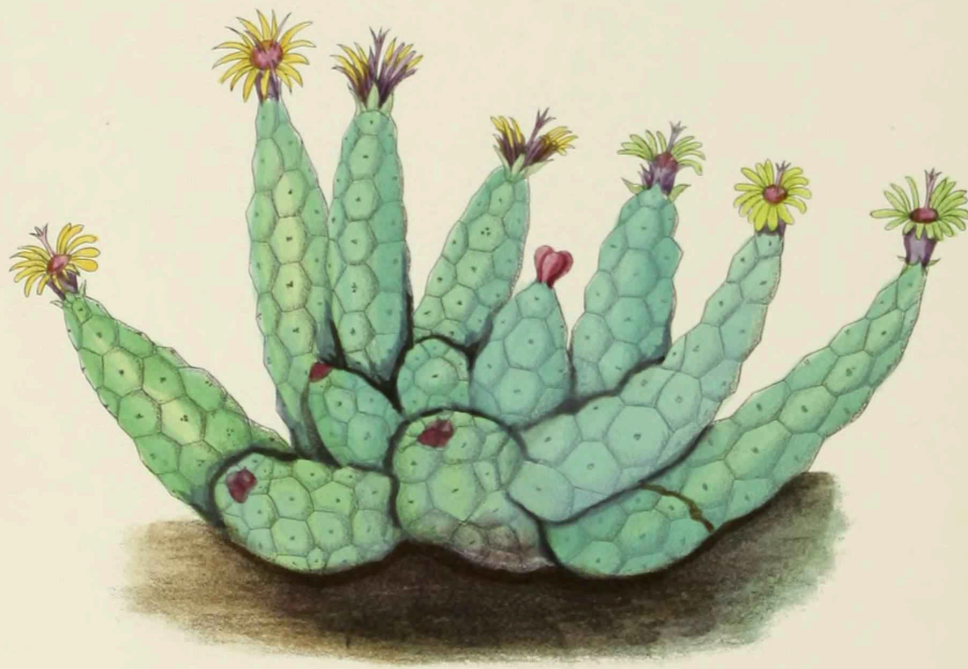


PLATE 196.

SUTERA BURKEANA.

Transvaal, Zululand.

SCROPHULARIACEAE. Tribe *Manuleae*.

SUTERA, *Roth.* (including *Lyperia*, Benth.); *Benth. et Hook. f. Gen. Plant.*
vol. ii. p. 945.

Sutera Burkeana, *Hiern. Fl. Cap.* vol. iv. sect. ii. p. 299.

The *Sutera Burkeana* here figured for the first time is a common plant in some localities around Pretoria and is to be found in flower from the beginning of August onwards. It is a bush up to 2-1/2 ft. high and appears to be very localised in its distribution, as only isolated patches are met with on the hills surrounding Pretoria. It has also been recorded from the neighbourhood of Johannesburg and from Zululand.

The plant flowers profusely, and when in full bloom makes a fine display and would no doubt be improved under proper cultivation.

Our Plate was prepared from specimens collected by Mr. C. A. Smith near Pretoria.

DESCRIPTION:—A much-branched shrub 0·3 to 1·3 m. high. *Branches* glandular-puberulous. *Leaves* fascicled recurved 3 to 8 mm. long, linear to oblong, cuneate at the base, acute, toothed, punctate. *Flowers* arranged racemosely at the ends of the branches. *Pedicels* 7 mm. long, terete, rigid, viscid-puberulous. *Calyx* 3 to 4 mm. long, deeply 5-lobed; lobes lanceolate-linear or oblong-lanceolate, acute, glandular-hairy. *Corolla-tube* cylindric, somewhat swollen and twice curved above, glandular-hairy without, softly hairy in the throat; lobes broadly obovate, obtuse, entire, often curled or recurved, glandular-hairy outside. *Stamens* 4, hardly exserted; filaments filiform, inserted on the corolla-tube. *Anthers* 1-theous, all perfect. *Ovary*

2-chambered with numerous ovules; style subulate, included; stigma obtuse. *Fruit* a capsule; valves cleft at the apex. *Seeds* numerous, regose. (National Herb., Pretoria, No. 2937.)

PLATE 196.—Fig. 1, longitudinal section of flower; Fig. 2, pistil; Fig. 3, cross-section of ovary.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 197.

EUPHORBIA TRIDENTATA.

Cape Province.

EUPHORBIACEAE. Tribe EUPHORBIEAE.

EUPHORBIA, *Linn.; Benth. et Hook. f. Gen. Plant.* vol. iii. p. 258.

Euphorbia tridentata, *Lam. Encycl.* ii. 416; *Fl. Cap.* vol. v. sect. ii. p. 298.

The species of *Euphorbia* figured on the accompanying Plate is a somewhat rare plant and until quite recently it was not known where the plant was found, although it was known in European gardens and figured almost 100 years ago. The plant belongs to a small group of three species in the genus characterised by having the branches constricted at their origin of growth. All of them are very dwarf plants.

We are indebted to Mr. H. M. Bartlett of Riversdale for the specimens which he sent in July 1924, and these were successfully grown at the Division of Botany, Pretoria, and flowered in September 1925.

DESCRIPTION:—Plant dwarf, succulent, spineless, branching from the base. *Branches* ascending or somewhat spreading, 2·5 to 15 cm. long, 1 to 1·4 cm. thick, cylindric or slightly tapering upwards, tessellately tuberculate with hexagonal flattish tubercles 6 to 10 mm. in diam., having a slightly prominent whitish leaf-scar, glabrous, dull green. *Leaves* sessile, soon deciduous, 4 to 6 mm. long, 3 to 4 mm. broad, elliptic or elliptic-oblong, acute, dark green, with a reddish minutely toothed margin. *Peduncles* 3 to 4 at the ends of the branches, about 4 mm. long, bearing a pair of ovate or elliptic bracts and 1 involucre, glabrous. *Involucre* about 1·3 to 1·8 cm. in diam., cup-shaped, glabrous, with 5 glands and 5 transversely oblong, toothed and ciliate, inflexed, purplish lobes. *Glands* subcontiguous, about 5 mm. in diam. across the tips, very concave at the basal part, divided into 3 to 4 spreading finger-like corrugated white processes 2 to 3 mm. long. *Ovary*

pedicellate, scarcely exserted, with styles 7 mm. long, united for two-thirds of their length, with entire spreading tips. (National Herb., Pretoria, No. 2989.)

PLATE 197.—Fig. 1, inflorescence; Fig. 2,
gland with lobes.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 198.

VENIDIUM WYLEYI.

Little Namaqualand.

COMPOSITAE. Tribe ARCTOTIDEAE.

VENIDIUM, *Less.; Benth. et Hook. f. Gen. Plant*, vol. ii. p. 459.

Venidium Wyleyi, *Harv. Fl. Cap.* vol. iii. p. 463.

This Plate, and that on Plate 117, should be compared with *Arctotis Fosteri* which we figured on Plate 3, when the difference between the genera *Venidium* and *Arctotis* will be seen. In the former genus the fruits are without a pappus or with a very rudimentary one, while in the latter genus a well-developed pappus is always present.

The plant is commonly known as the “Namaqua Daisy,” and in its native home the flower-heads attain a very large size when the season is favourable, but may be quite small when no rain has fallen. *Venidium Wyleyi* does quite well under cultivation, and is easily grown and forms a very effective border.

We are indebted to Mr. L. R. Vogt of Waterkloof near Pretoria for the specimens, which were grown in his garden.

DESCRIPTION:—An erect herb. *Stems* terete, hollow, striate, pilose-hoary, laxly leafy. *Cauline leaves* 1·2 to 15 cm. long, sessile; the lower lyrato-pinnatifid, amply auricled, amplexicaul, on both sides cobwebby-tomentose; the upper broadly ovate, closely crowded in the young stems, cobwebby-tomentose, obtuse, with the margins entire or remotely 1 to 2 toothed. *Heads* peduncled, terminal, very large and showy, 8 cm. in diam. (including the rays), the buds very hoary. *Involucral bracts* imbricate, in several rows; the outer narrower, reflexed, herbaceous, green; the innermost row much longer, broadly scariose, spreading below the rays; the whole involucre densely cobwebby. *Receptacle* 1·8 cm. in diam., convex, deeply honeycombed, with

the margins of the cells produced into a few bristles. *Ray-florets* female, 1-seriate, alternate rays spreading, others at first ascending, then spreading, giving an appearance of two rows. *Corolla* ligulate; lamina bright orange coloured, oblong-cuneate, very faintly 3-toothed at apex, of lower whorl smaller (2·5 cm. long) and with a smaller brown-black blotch at the base than that of upper whorl (3 cm. long); tube 3 mm. long, broader at base than at the top, subglabrous. *Ovary* obovate in outline, muricated, glabrous; style 4 mm. long, cylindric, glabrous; stigmatic portion wider, with the branches 0·5 mm. long, linear, flat, spreading. *Disc-florets* hermaphrodite, fertile, numerous, crowded on the head. *Corolla* 4 mm. long, subcylindric, 5-cleft at the apex for about a quarter the length of the tube, with 5 longitudinal rows of glandular hairs; lobes blackish, 1 mm. long, linear, subacute, with the angles between the lobes obtuse. *Anthers* purplish, black, 2 mm. long, obtuse, with a small suborbicular apical appendage; filaments 1·5 mm. long, linear, flat, at length filiform, inserted on lower half of corolla tube. *Ovary* obovate in outline, white, muricated, glabrous; style 5·5 mm. long, at first narrowly cylindric, then in upper half abruptly widening into a yellow stigmatic portion, 2 mm. long, slightly cleft at the apex, becoming much exserted. (National Herb., Pretoria, No. 2944.)

PLATE 198.—Fig. 1, longitudinal section of receptacle; Fig. 2, ray-floret; Fig. 3, disc-floret; Fig. 4, inner involucral-bract; Fig. 5, outer involucral-bract; Fig. 6, achene.

F.P.S.A., 1923.



K.A. Lansdell del.

PLATE 199.

ERICA BLENNA var. GRANDIFLORA.

Cape Province.

ERICACEAE. Tribe ERICEAE.

ERICA, *Linn.*; *Benth. et Hook. f. Gen. Plant.* vol. ii. p. 590.

Erica blenna, *Salisb.* var. **grandiflora**, *Bolus. Fl. Cap.* vol. iv. sect. i.
p. 202.

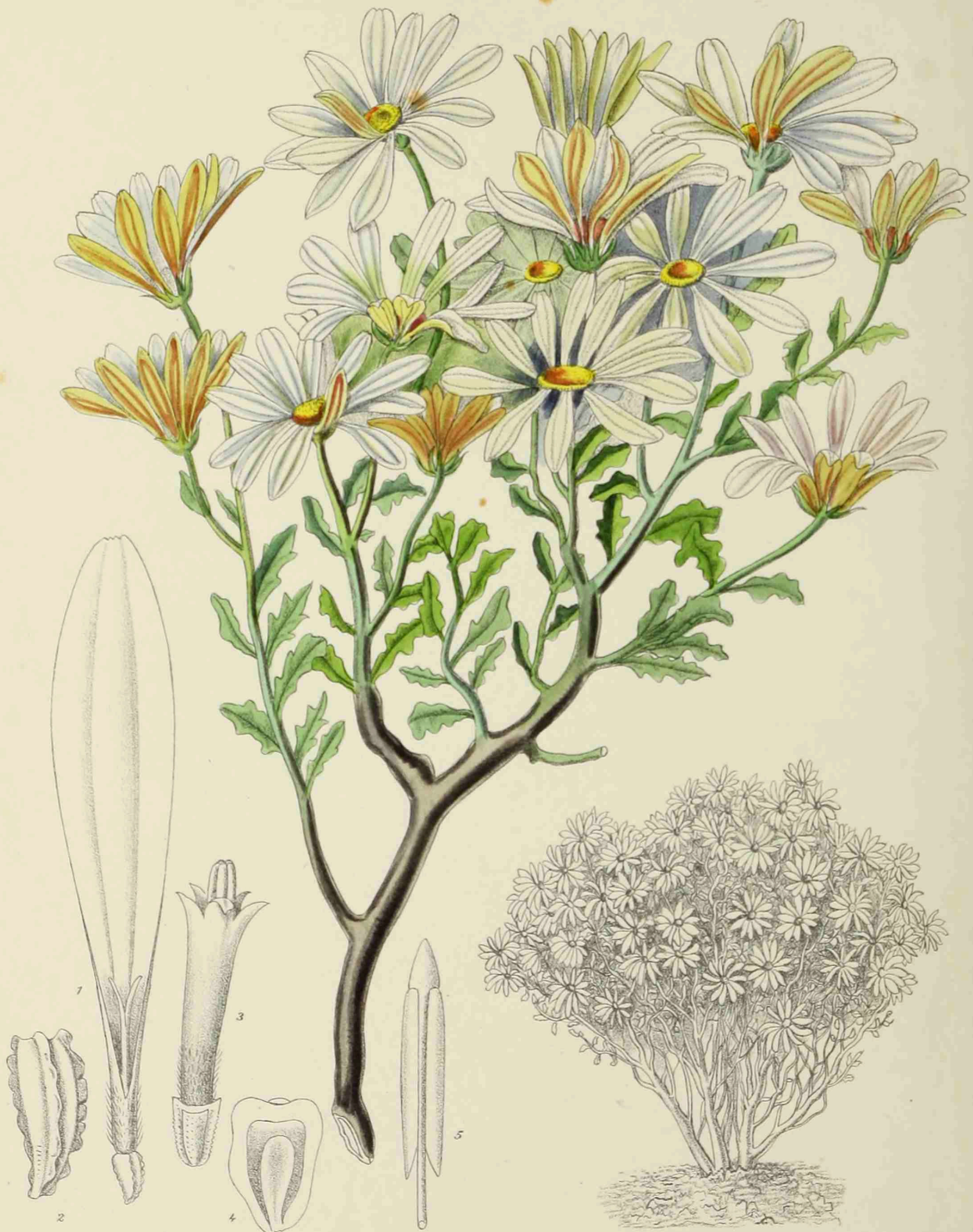
Through the courtesy of Dr. J. Muir of Riversdale we are able to figure for the first time a species of heath, and perhaps one of the most beautiful of all the Cape heaths. The variety *grandiflora* is only found as far as we know in the Riversdale and Bredasdorp Districts and is by no means a common plant. It is known locally as the “lantern” or “Riversdale” heath. The species itself, like so many of the South African species of *Erica*, was first collected by Masson about the year 1772, and was successfully cultivated in Europe and figured on more than one occasion in early botanical literature.

DESCRIPTION:—Erect, 0·3 to 0·5 m. high. *Branches* stout, ascending, virgate or flexuous, puberulous or glabrous. *Leaves* 3-nate, mostly erect and imbricate or subspreading, linear, subobtuse, flat above, keeled and sulcate beneath, glabrous, 8 to 10 mm. long. *Flowers* usually solitary, rarely in pairs (“here and there sublateral,” Bentham). *Pedicels* about 8 mm. long; bracts remote, lanceolate, about 6 mm. long; sepals ovate, acuminate, keel-tipped, thickish, subscarious, viscid, coloured or greenish, about 5 mm. long. *Corolla* conical-ovoid or suburceolate-conical, much contracted to the mouth but only slightly constricted at the throat, very viscid, 1·6 to 2 cm. long, bright orange-red, the limb and some distance below it green; segments spreading or erect, about one-eighth the length of the tube; filaments broad at the base tapering upwards, bent below the anther; anthers included, dorsifixed well above the base, cuneate, subacute, scaberulous, ciliolate,

about 3 mm. long, crested; pore three-fifths to two-thirds the length of the cell; crests quite free from the filament, subsemiorbicular in outline, deeply inciso-lacerate, about half the length of the cell; style included, straight; stigma capitellate; ovary glabrous. (National Herb., Pretoria, No. 2991.)

PLATE 199.—Fig. 1, pedicel showing bracts; Fig. 2, androecium and gynaecium; Fig. 3, single stamen; Fig. 4, pistil; Fig. 5, anther enlarged.

F.P.S.A., 1925.



K.A. Lansdell del.

PLATE 200.

DIMORPHOTHECA CUNEATA.

Cape Province, Natal, Orange Free State.

COMPOSITAE. Tribe CALENDULEAE.

DIMORPHOTHECA, *Moench.*; *Benth. et Hook. f. Gen. Plant.* vol. ii. p. 453.

Dimorphotheca cuneata, *Less. Syn.* 257; *Fl. Cap.* vol. iii. p. 422.

We are indebted for the specimens, from which the accompanying Plate was prepared, to Mr. C. A. Smith, B.Sc., who collected them in September 1925 on the botanical reserve near Fauresmith, Orange Free State. According to Mr. Smith the plants cover the hill-sides and from a distance appear as large white patches. It flowers very profusely, so much so that the leaves are almost hidden by the mass of flowers. The laminae of the ray-florets are white above and copper-coloured beneath, not yellow as described in the *Flora Capensis*.

The specimens collected by Mr. Smith are an exact match with those collected by Zeyher (No. 2812). Zeyher No. 3066, quoted by Harvey as *D. cuneata*, is quite a different plant, perhaps a *Tripteris*. We have accepted the specific name "cuneata," as this was the name given by Mr. N. E. Brown to a specimen collected by Mr. E. E. Galpin. Locally known as "mak-bietou."

DESCRIPTION:—A densely and closely branched bushy shrub up to 1·05 m. high, the older branches naked, rough, the younger short and closely leafy. *Leaves* varying in shape (linear to obovate) and size (0·5 to 1·3 cm. long), always cuneate at the base, sharply 2 to 4 toothed, obscurely mid-ribbed, gland-dotted, subdecurent, glabrous or nearly so. *Peduncles* terminal, 2 to 3·5 cm. long, glandular-pubescent, viscidulous, becoming widened at the top. *Involucre* uni-seriate, glaucescent, viscidulous; scales linear-acuminate, minutely glandular-puberulous, with pale-edged and ciliate margins. *Receptacle* about 3·5 mm. in diam., nude, flat. Ray-florets female,

uniseriate. *Corolla* ligulate, white above, yellow to bronze-coppery below; lamina spreading, finely-three toothed, cuneate at the base; tube of corolla 1·5 mm. long, terete, glandular-hairy. *Ovary* triquetrous, green, obconic, somewhat curved, glandular hairy with stalked glands, with the angular margins bluntly toothed; style 4·5 mm. long, cylindric, glabrous; branches 2 mm. long, yellow, flat, subacute. *Disc-florets* hermaphrodite, fertile. *Corolla-tube* yellow, 3 mm. long, subcylindric, with a 5-fid limb, very densely glandular-hairy at the base, glabrous above. *Anthers* 3 mm. long, subsagittate at the base, with an ovate obtuse apical appendage. *Ovary* very much laterally compressed, obcordate, glandular, with a wide thick-rimmed entire marginal wing; style 6 mm. long, cylindric, glabrous; branches appearing truncate with a ring of bristles, capped by a short conical apex. (National Herb., Pretoria, No. 2990.)

PLATE 200.—Fig. 1, ray-floret; Fig. 2, ray-achene; Fig. 3, disc-floret; Fig. 4, disc-achene; Fig. 5, stamen.

F.P.S.A., 1925.

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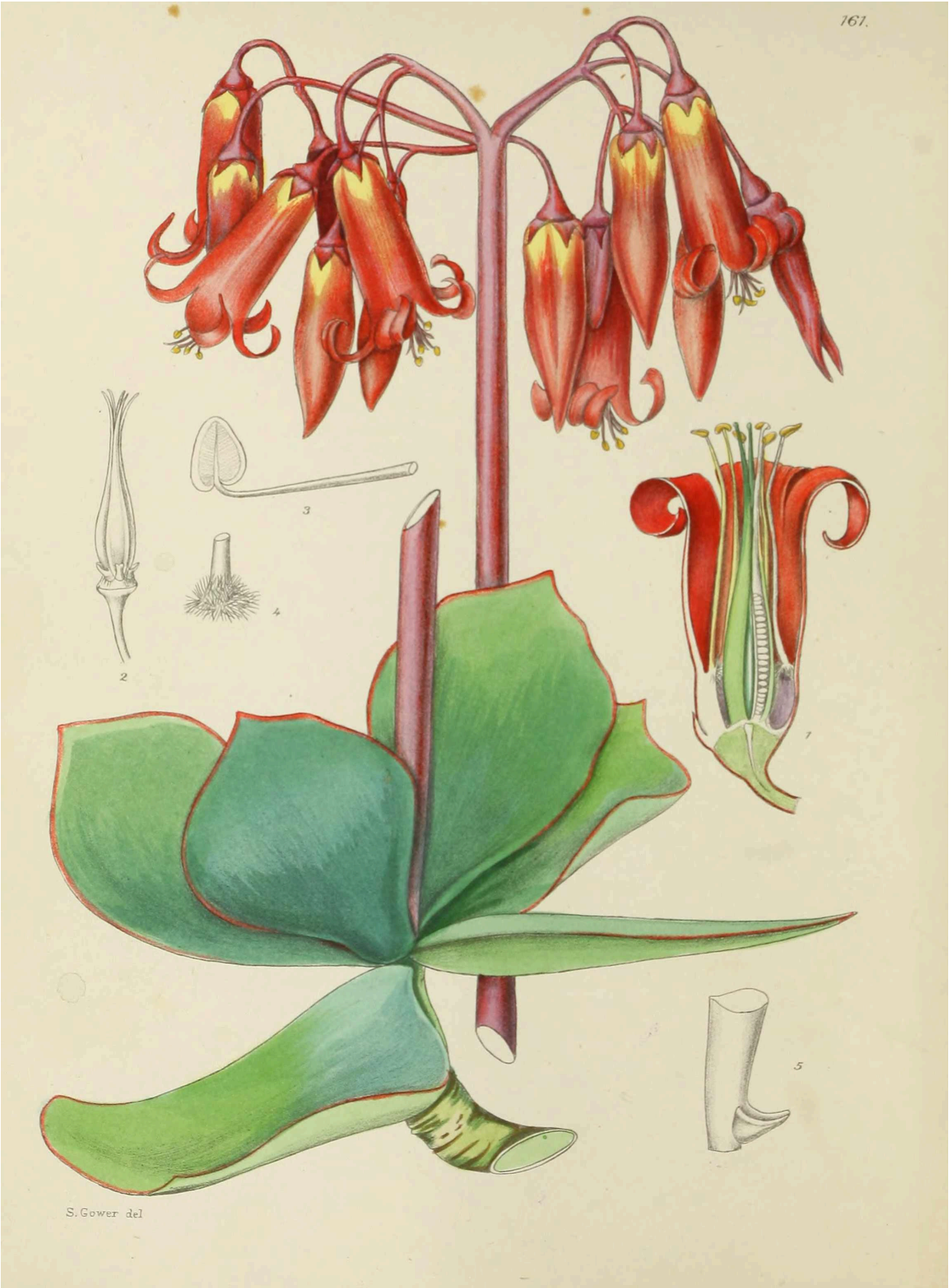
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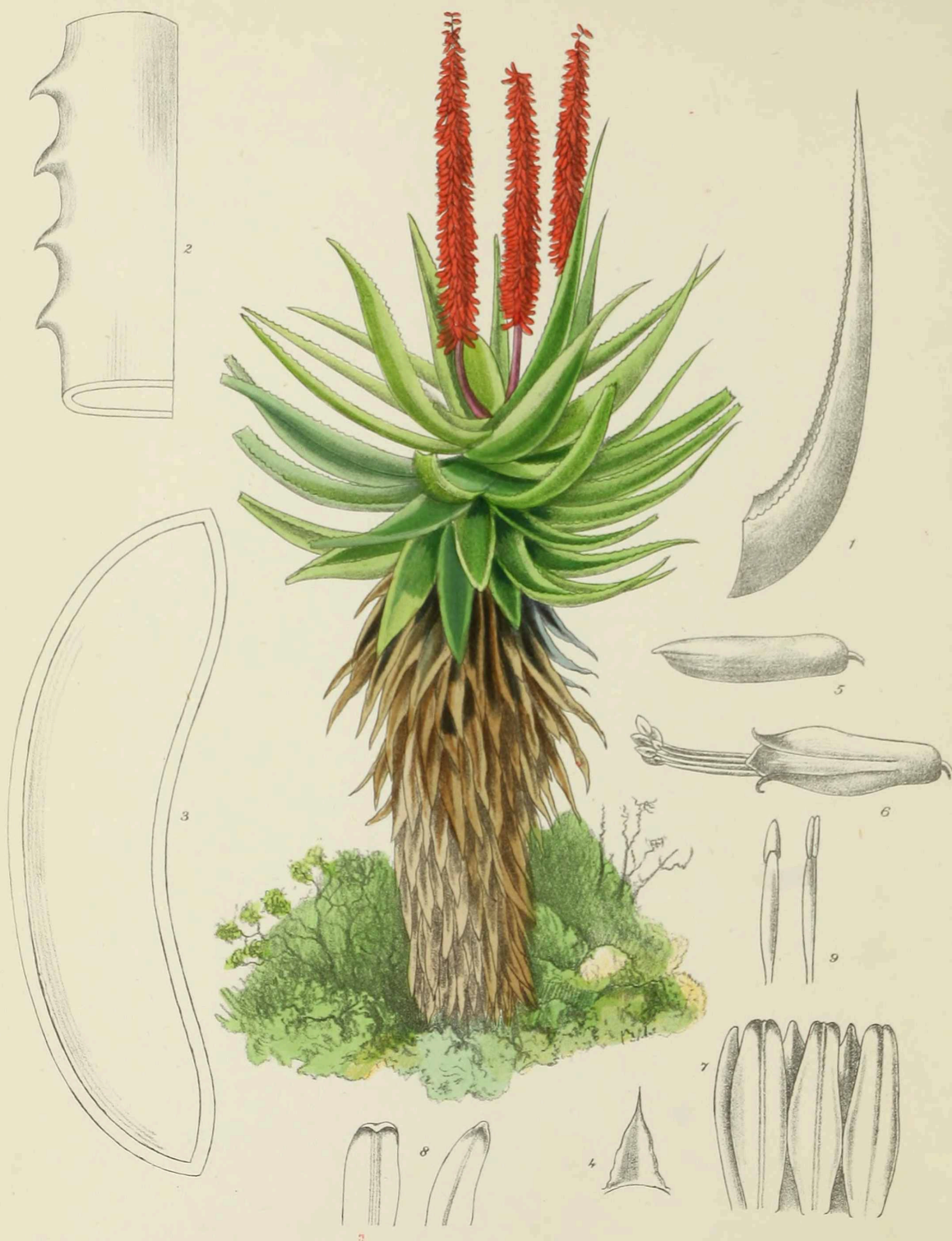
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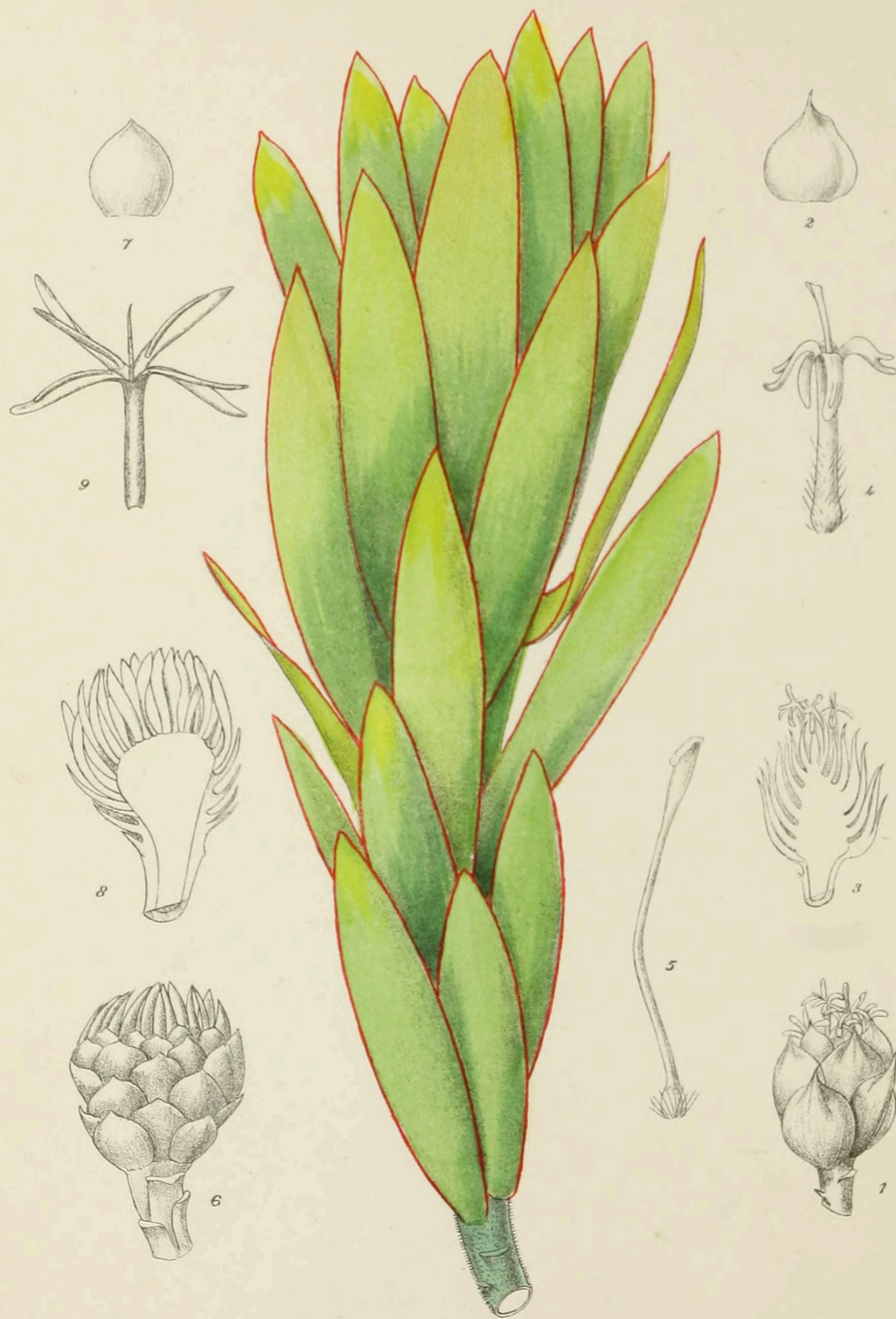
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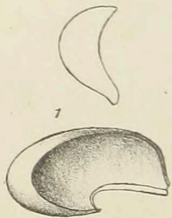
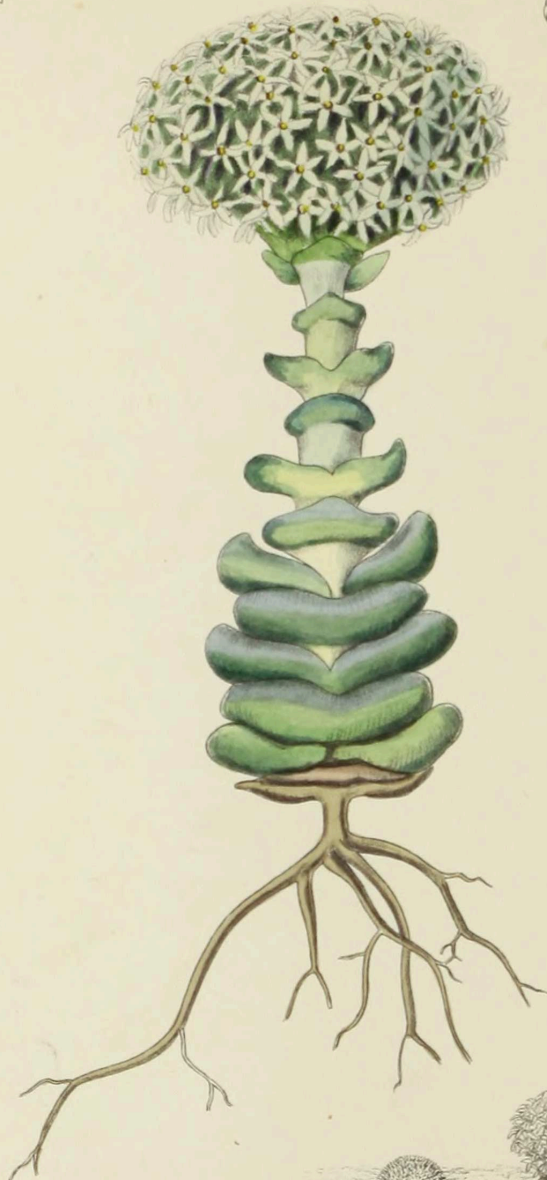
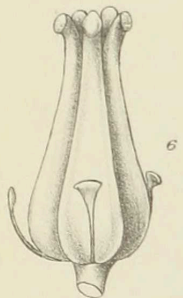
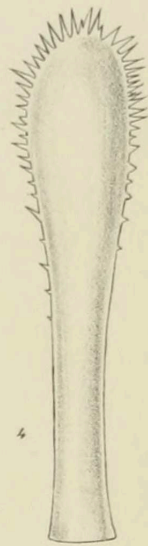
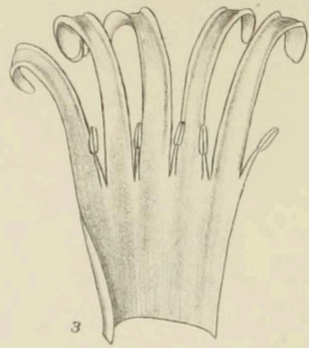
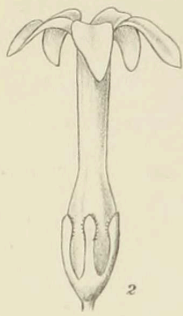
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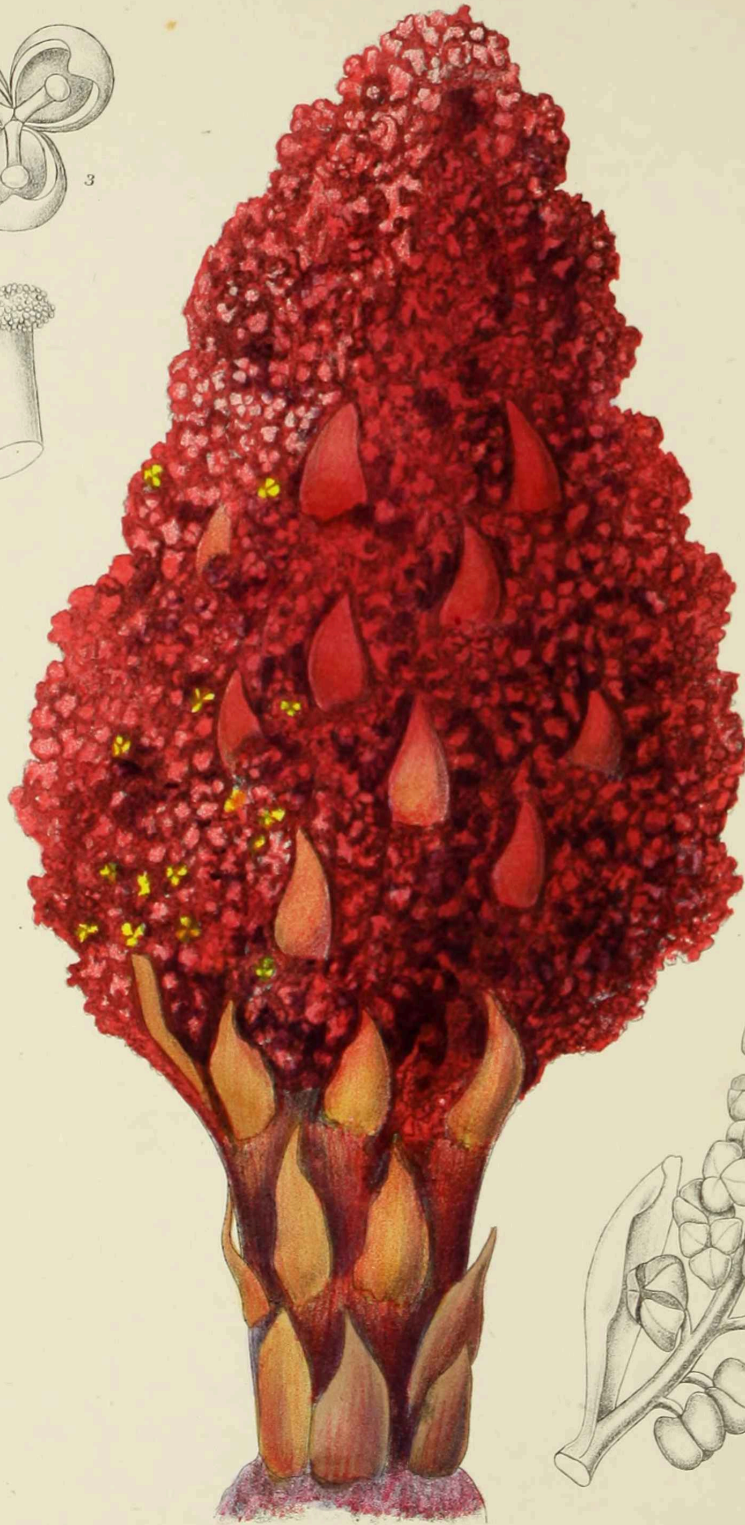
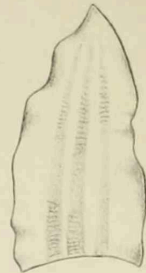
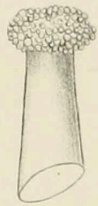
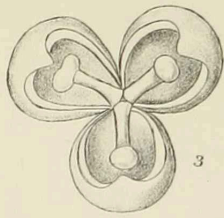


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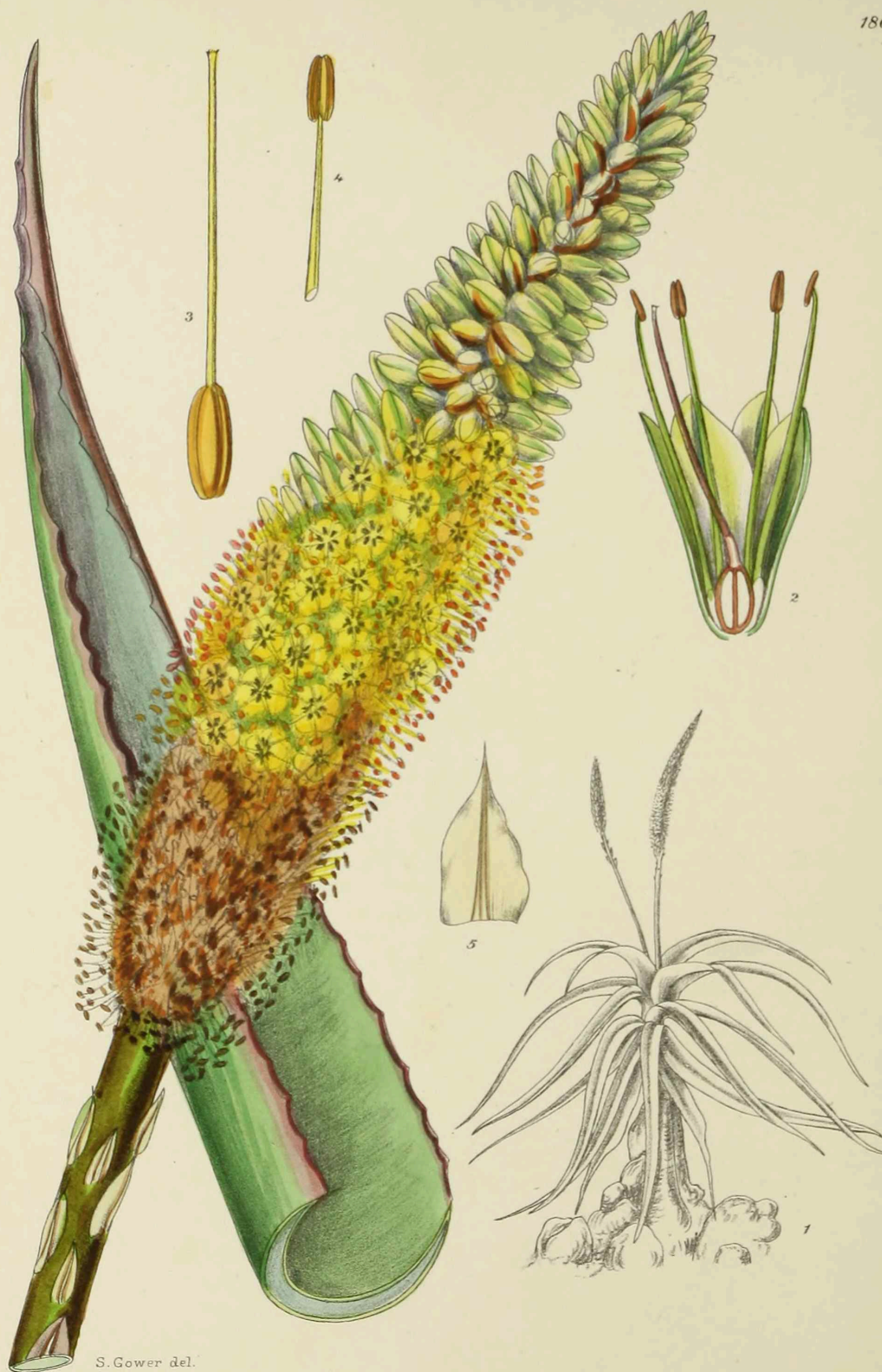


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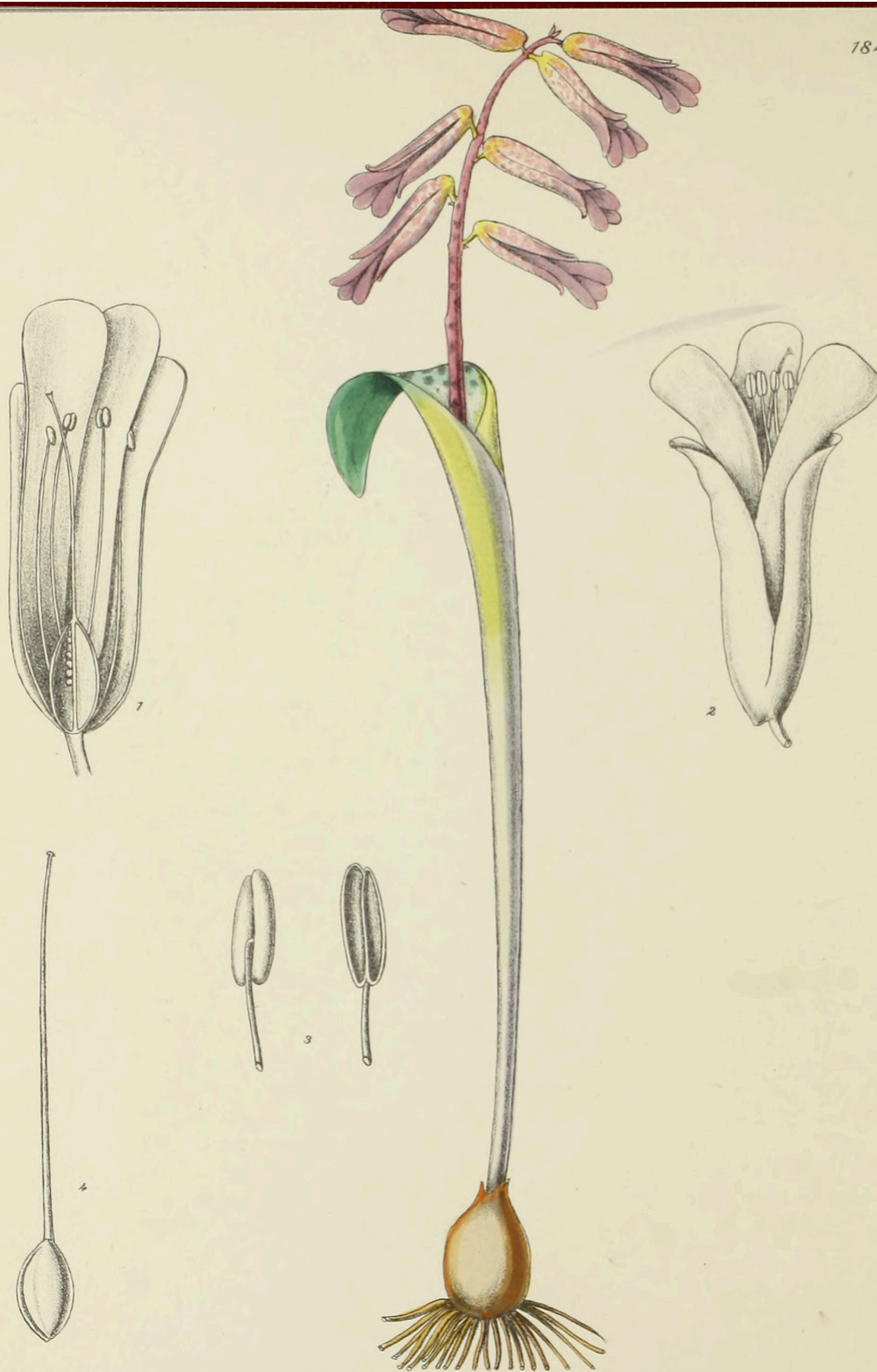
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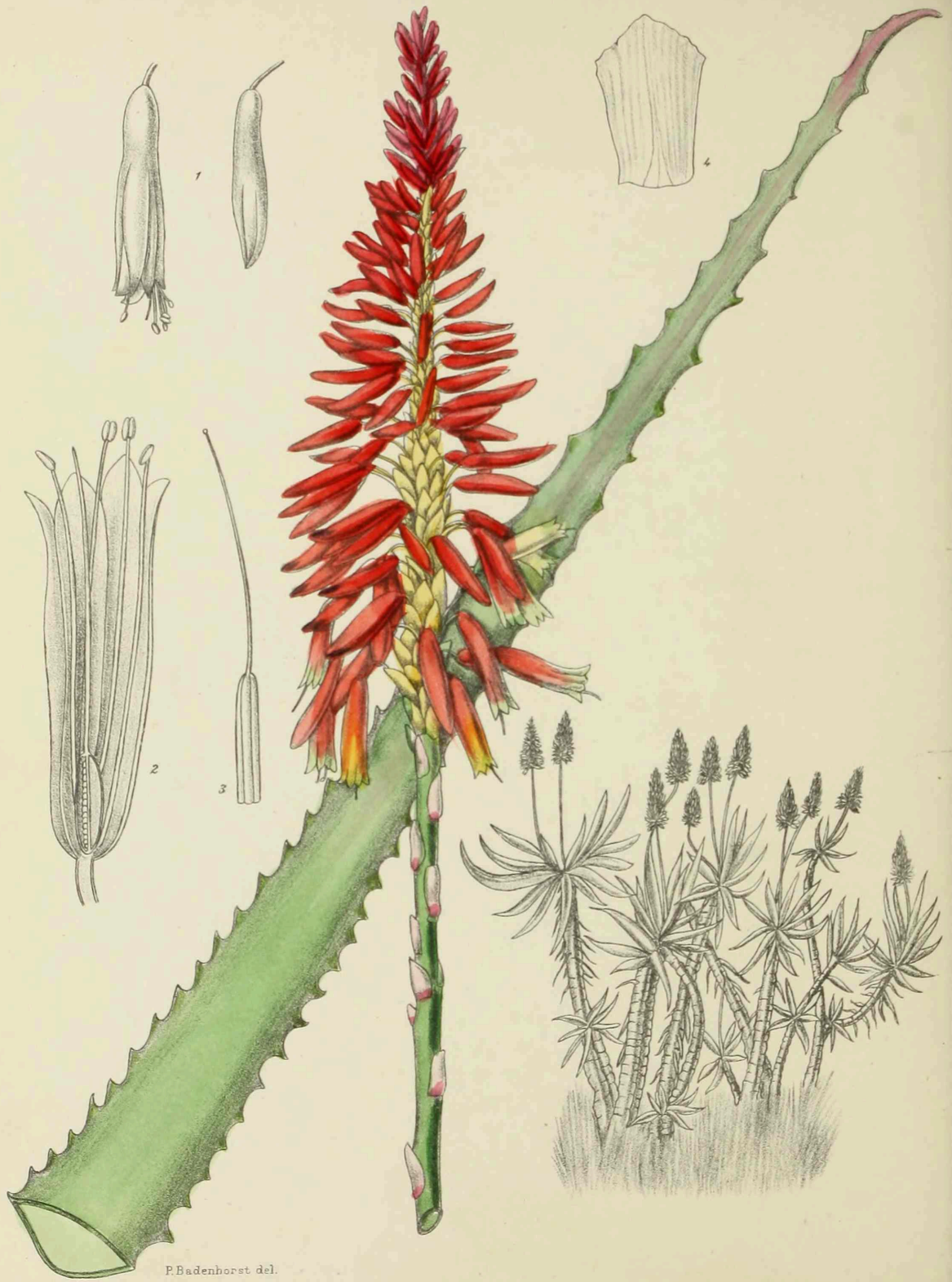
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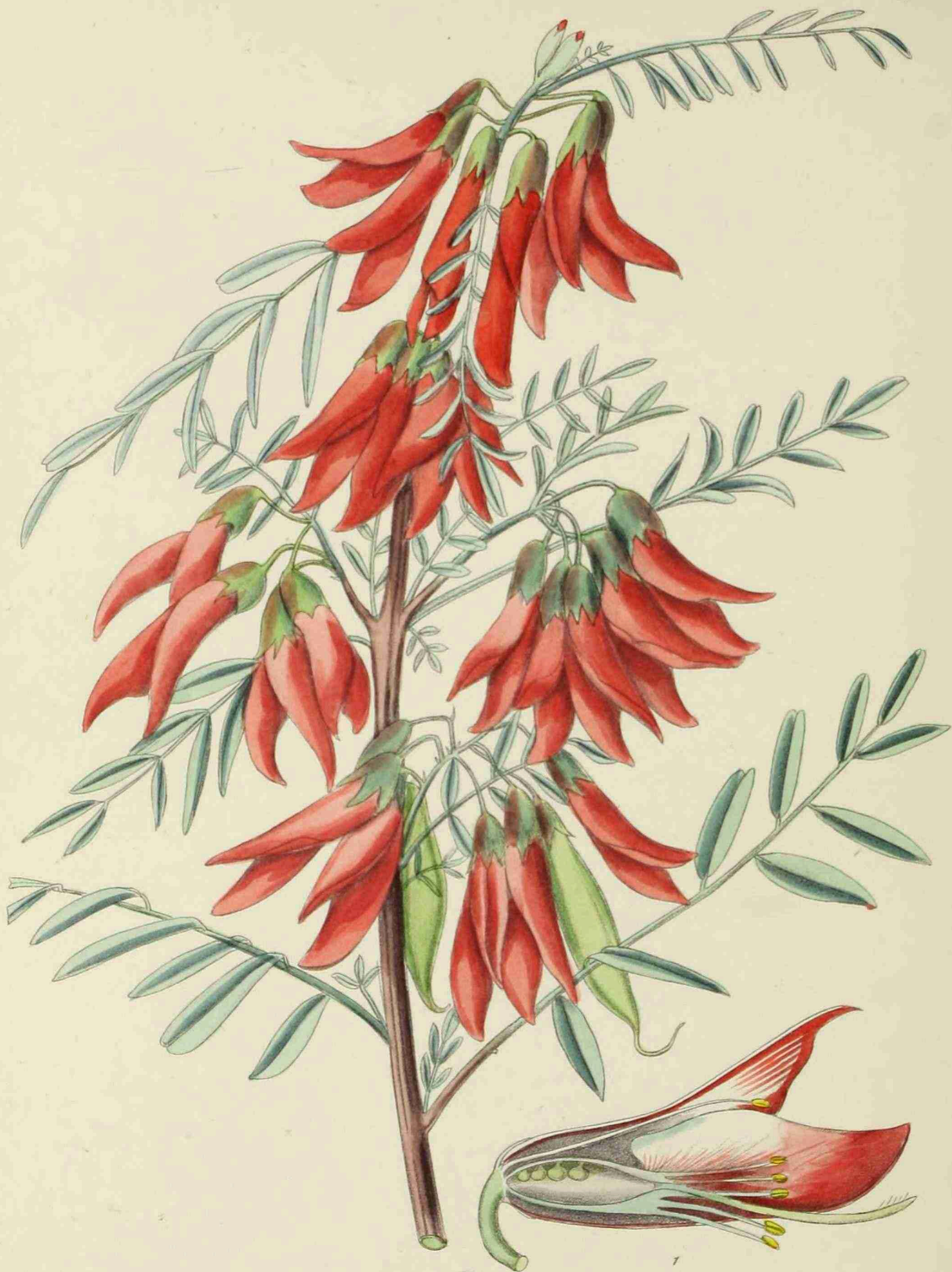


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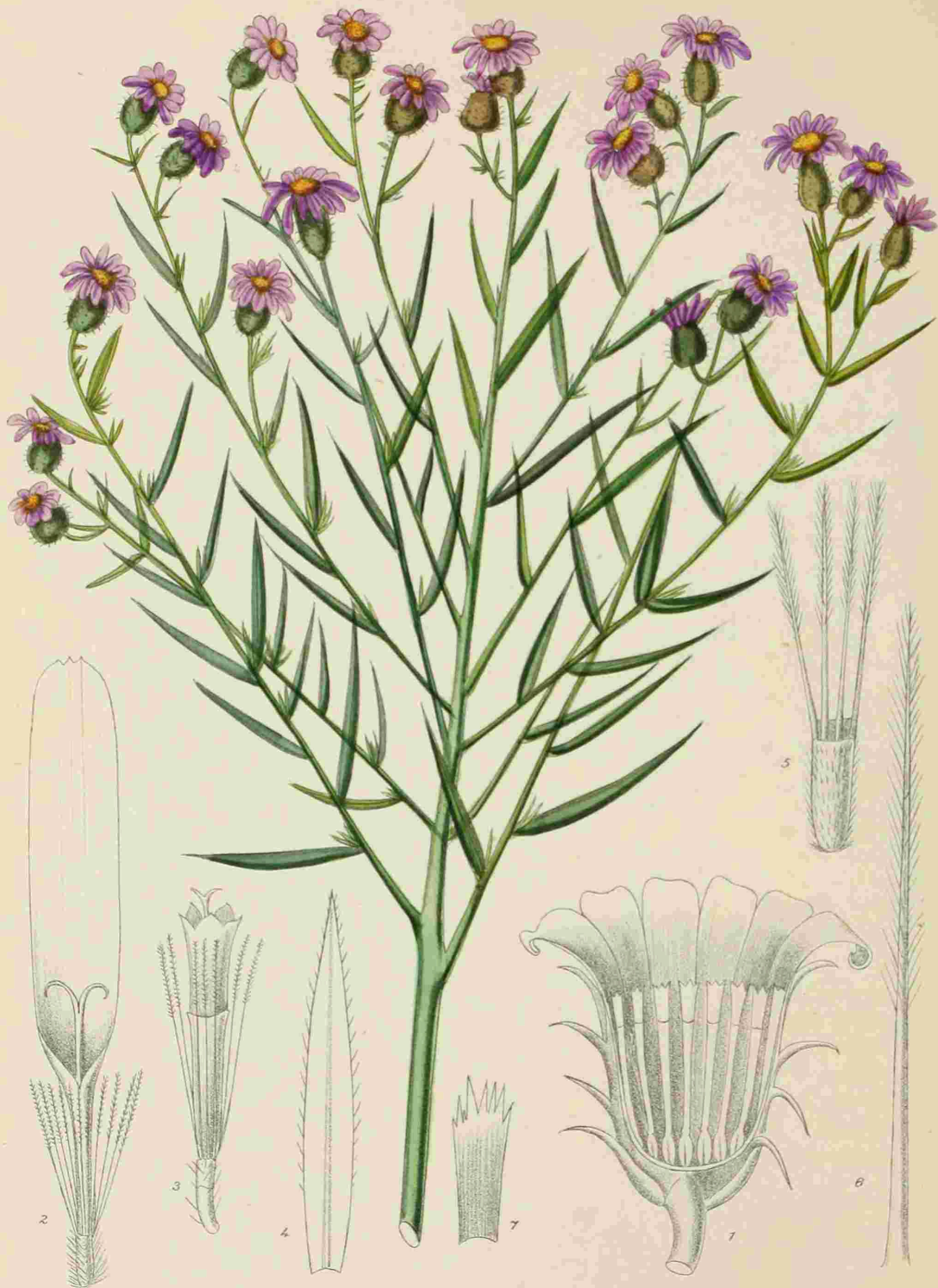


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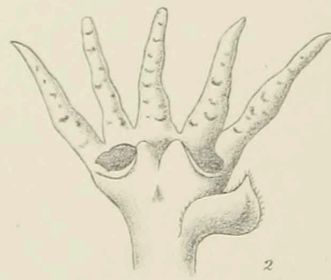
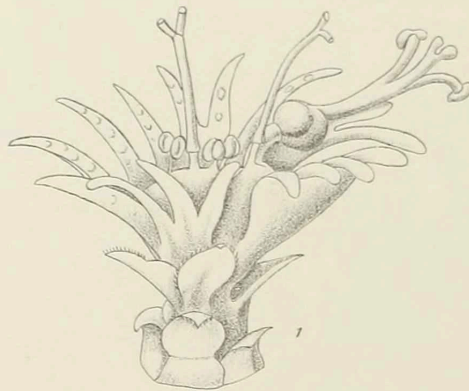
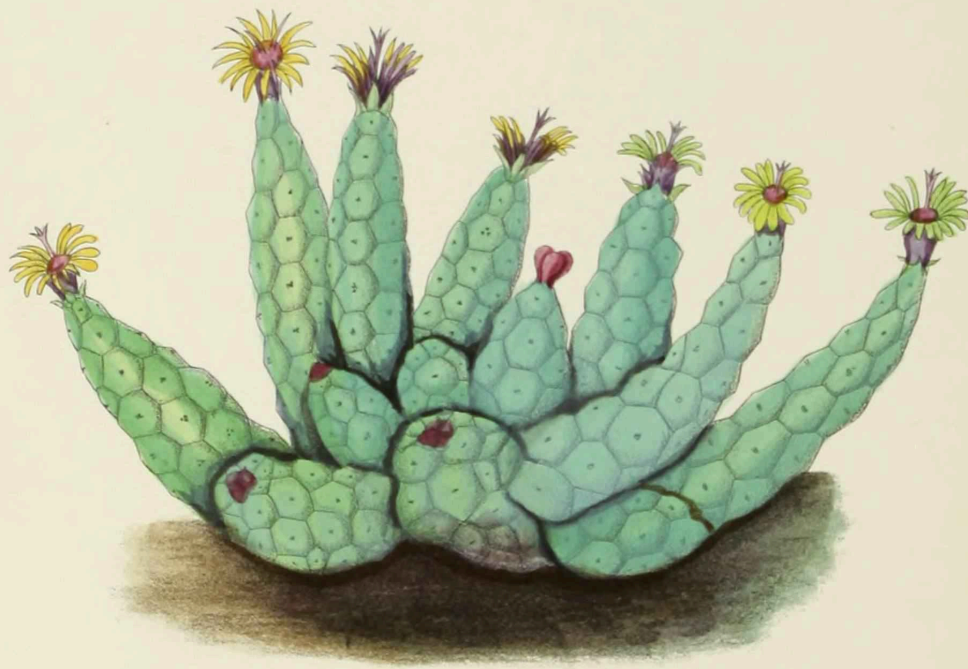


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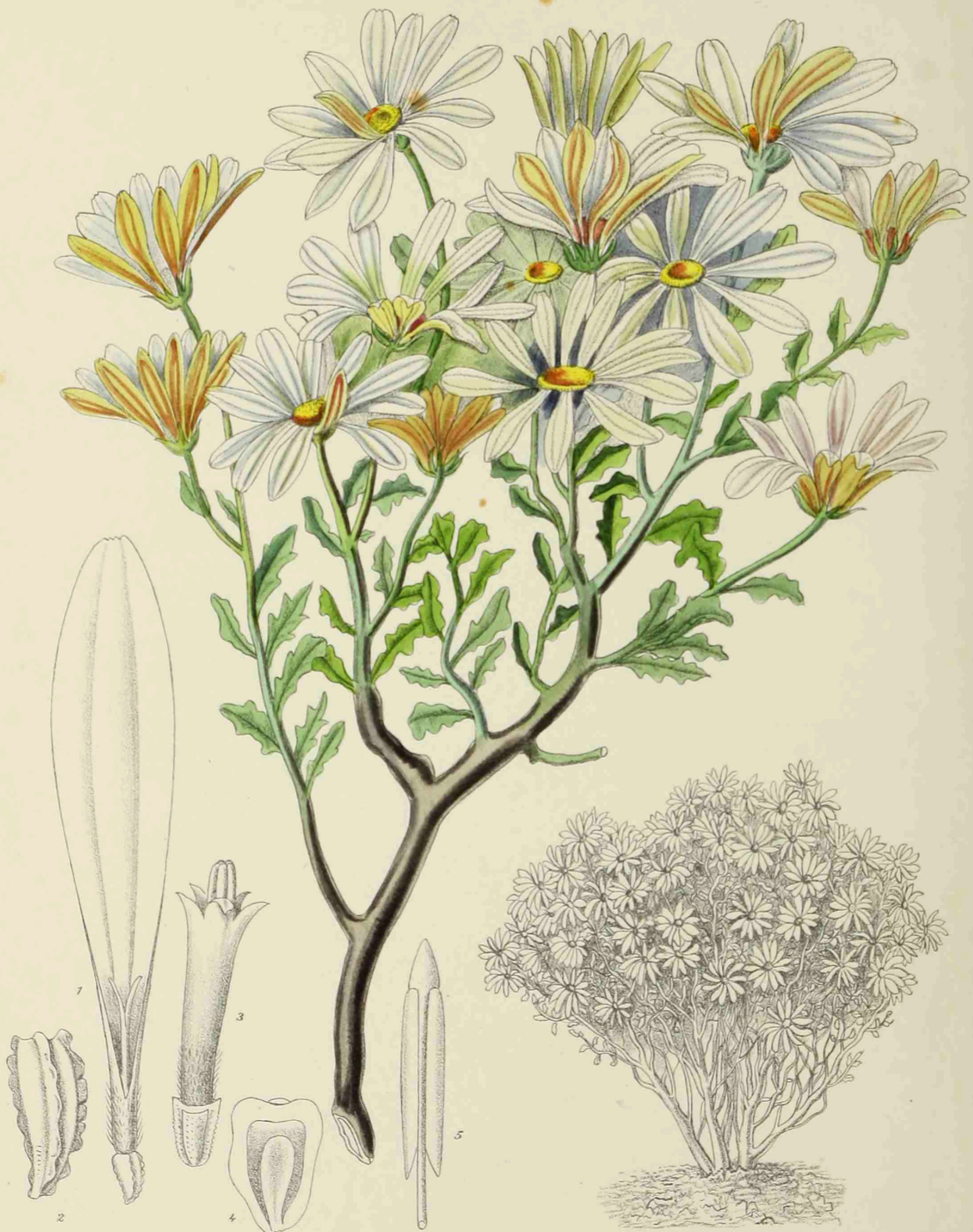
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